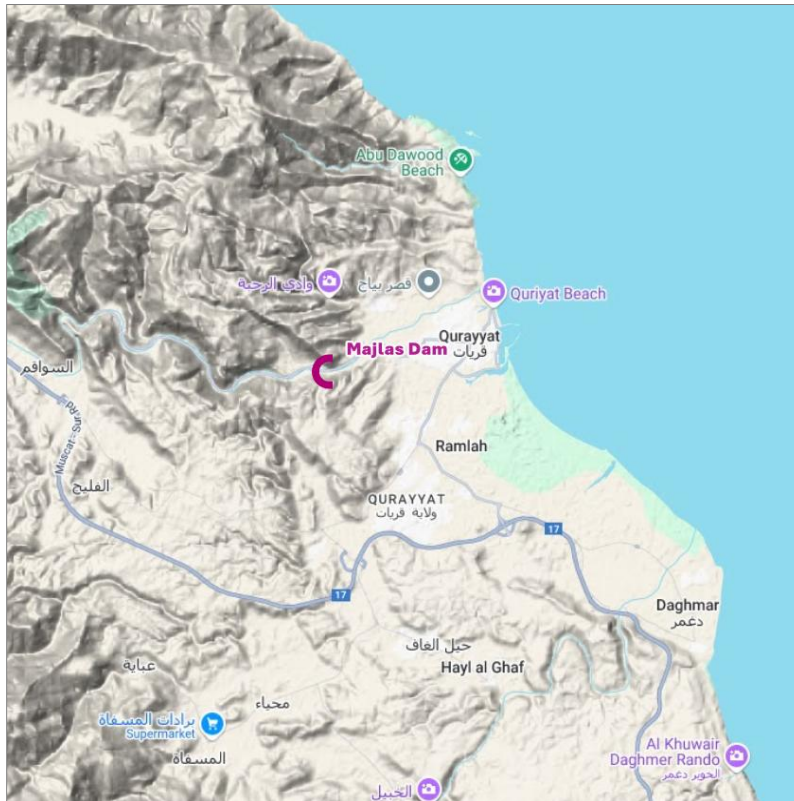




Sultanate of Oman
Ministry of Agriculture, Fisheries Wealth & Water Resources

Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate



Wadi Majlas Flood Protection Dam Dam Break Analysis Report

Rev 01
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ABBREVIATIONS AND NOMENCLATURE

Terms and short names used in this report.

Table 1 - Abbreviations and terms.

Term	Meaning
AIDR	Australian Institute for Disaster Resilience; its flood hazard classes H1 to H6 are used here
Breach	The opening that forms in the dam when it fails
EAP	Emergency action plan
FSL	Full supply level, 84.5 m a.s.l., the spillway crest
HEC-RAS	The two-dimensional hydraulic model used for the analysis (Hydrologic Engineering Center, River Analysis System)
Monolith	One concrete block of the dam between two vertical joints; the dam has 17 of them
PAR	Population at risk: people living in the flooded area
PMF, PMP	Probable maximum flood, probable maximum precipitation
S1	Sunny-day failure: the dam fails with the reservoir full and no flood in the wadi (S1D with the dikes, S1N without)
S2	Flood-day failure: the dam fails at the peak of the PMF (S2D)
S3	PMF without failure: the flood alone, the reference for what the failure adds (S3D)
D, N	With the downstream training dikes; without them
S1D-W51, S1D-W153, S2D-W119	Sensitivity runs with a breach 51, 153 or 119 m wide instead of 85 m
S1D-T3m, S1D-T18m	Sensitivity runs with the breach formed in 3 or 18 minutes instead of 6
S1D-C1.44	Sensitivity run with a breach weir coefficient of 1.44 instead of 1.66
S2D-F10000, S3D-F10000	The flood-day failure and the no-failure run repeated with the 10,000-year flood instead of the PMF
m a.s.l.	Metres above sea level



EXECUTIVE SUMMARY

This report presents the dam break analysis of the Wadi Majlas Flood Protection Dam, a 72 m high roller-compacted concrete gravity dam about 2 km upstream of Qurayat. A two-dimensional hydraulic model of the reservoir, the gorge and the coastal plain down to the sea was built for the study. Three scenarios were analysed: a sunny-day failure (S1D), a failure at the peak of the probable maximum flood (S2D) and the probable maximum flood without failure (S3D), the last being the reference against which the failure is measured. The sunny-day failure was also run without the downstream training dikes (S1N), and eight sensitivity runs bound the breach parameters and the flood: twelve result runs in all.

Sunny-day failure (S1D). With the reservoir at the full supply level and no flood, the sudden failure of five monoliths (85 m of the dam) releases 103 Mm^3 in about two hours. The peak outflow is $45,312 \text{ m}^3/\text{s}$, reached 10 min after the failure begins. Water deeper than 0.3 m covers 17.0 km^2 of the coastal plain. The wave reaches the plain within 15 min, the residential centre of Qurayat within 15 min and the shoreline within 22 min. About 23,212 people live in the flooded area, 22,541 of them where the flood is unsafe for people and vehicles or worse; 15,579 of them are reached within 15 minutes of the breach and 21,739 within 30 minutes.

Flood-day failure (S2D). The dam is assumed to fail when the reservoir reaches its maximum level during the PMF, 96.06 m a.s.l. at 15 h 15 min into the storm, a level and a moment taken from the no-failure run S3D. The peak outflow rises from $17,323 \text{ m}^3/\text{s}$ without failure to $63,858 \text{ m}^3/\text{s}$ with failure. The flooded area grows only from 25.9 to 27.1 km^2 , because the flood has already covered the plain; what the failure adds is depth, force and speed. The area in the worst hazard class grows from 13.0 to 18.4 km^2 , the people in the flooded area from 44,088 to 46,299, and the failure wave adds 0.3 m or more to the water already standing on 25,365 people within 15 minutes of the breach.

PMF without failure (S3D). The reservoir peaks at 96.06 m a.s.l., 0.64 m below the parapet, and the spillway passes $17,323 \text{ m}^3/\text{s}$ against its design discharge of $17,000 \text{ m}^3/\text{s}$. The dam is not overtopped. The flood alone covers 25.9 km^2 of the plain with 44,088 people in it, but it builds up over hours: the residential centre is reached 1 h 06 min after the storm starts and the shoreline 1 h 38 min after.

Effect of the training dikes. The dikes were designed for the 200-year wadi flood, not for a dam break. In the sunny-day failure the flooded area is 17.0 km^2 with the dikes and 17.1 km^2 without, and



the population at risk 23,212 and 23,299 respectively: a difference well under one per cent. The dikes were therefore kept in every other run and the no-dikes condition was not pursued further.

Warning time is what separates the scenarios. In extent and hazard class the flood without failure looks almost as bad as the failure; in time it does not. The sunny-day wave reaches 15,579 people within 15 minutes of the breach and 23,179 within an hour, whereas the flood without failure needs 1 h 06 min to reach the town and about twelve hours to reach everyone it will reach. Warning for a dam failure must therefore come from the dam itself, before the failure, not from the flood.

Sensitivity. The breach width sets the peak almost in proportion (31,663 to 66,955 m³/s for three to nine monoliths) but hardly moves the flooded area or the people in it; the formation time and the weir coefficient change the peak by about a tenth; the 10,000-year flood gives the same picture as the PMF with a tenth less flow. The base-case results are therefore robust, and the bounds are quoted with them.

The results are meant for the emergency action plan and for the design review: the arrival times and hazard maps show where warning and evacuation are needed first, and the no-failure runs confirm the spillway capacity against the PMF. Figure 1 puts the three scenarios side by side.

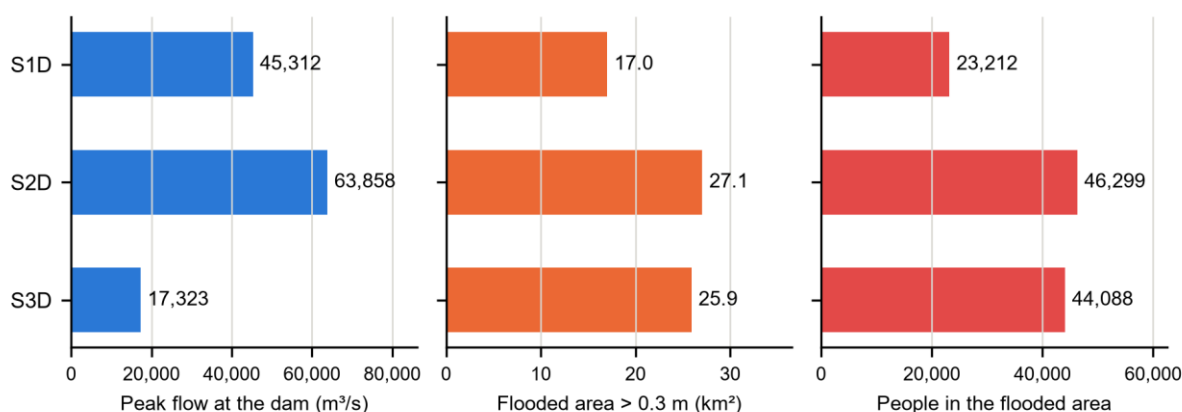


Figure 1 - The three scenarios at a glance: peak flow at the dam, flooded area and people in it.



1. INTRODUCTION

1.1. PURPOSE AND SCOPE

A dam break analysis answers three questions: how fast and how far would the water go if the dam failed, who and what would be in its way, and how much worse is a failure than the flood the dam is already passing. This report answers them for the Wadi Majlas dam with a two-dimensional hydraulic model, and gives the numbers that the emergency action plan¹ needs.

Chapter 2 sets out the method: how a concrete gravity dam fails, how the breach was sized, how the flood is routed and how the consequences are measured. Chapter 3 defines the scenarios and the runs, chapter 4 describes the model as built with every input, chapters 5 to 7 give the results of the sunny-day failure, the flood-day failure and the flood without failure, chapter 8 the sensitivity of the results to the breach parameters, chapter 9 the consequences for people and property, including who is reached and when. Chapters 10 and 11 discuss the findings and draw the conclusions.

1.2. THE DAM AND THE VALLEY IN BRIEF

The dam is a roller-compacted concrete (RCC) gravity dam with a grout-enriched RCC facing, 72 m above the wadi bed, 312 m long at the crest (95.5 m a.s.l.), with a 1.2 m parapet to 96.7 m a.s.l. The ungated ogee spillway, 200 m net length at 84.5 m a.s.l., is designed for 17,000 m³/s. The reservoir holds 102.2 Mm³ at the full supply level. Downstream, the wadi leaves a gorge and crosses the coastal plain of Qurayat, where two training dikes, 2.6 km long on each bank, guide the 200-year flood to the sea. Chapter 4 gives the dam as designed and the model built on it.

¹ Emergency action plan (EAP): the document that tells the operator and the authorities what to do, and whom to warn, when the dam is in danger. The arrival times and hazard maps in this report are its main technical input.



2. METHODOLOGY

2.1. THE ANALYSIS IN OUTLINE

The analysis runs from the data to the inputs of the emergency plan in the steps of Figure 2: the dam type fixes the failure mode and the breach; the scenarios fix the water in the reservoir and in the wadi at the moment of failure; the two-dimensional model routes the flood over the terrain; the results are turned into depth, velocity, hazard class and arrival time on a 2 m grid; and the people and properties in the flooded area are counted, by hazard class and by the time at which the water reaches them. Each step is described below.



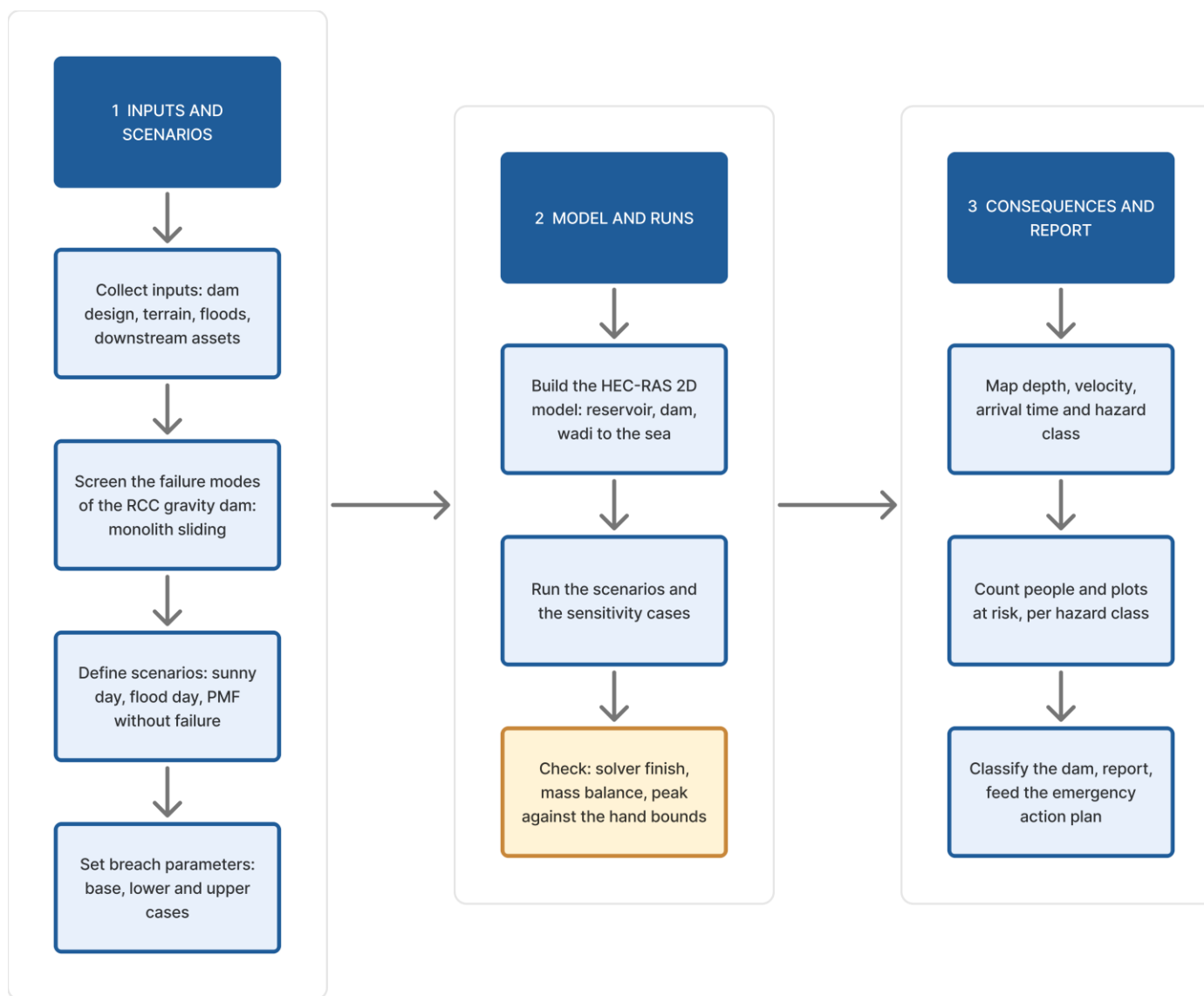


Figure 2 - The dam break analysis process, from data to emergency-plan inputs.



2.2. FAILURE MODE OF A CONCRETE GRAVITY DAM AND THE BREACH PARAMETERS

A roller-compacted concrete gravity dam does not erode. It fails, if at all, by the sudden sliding or overturning of one or more monoliths, the blocks between vertical contraction joints, when the water load exceeds what the foundation contact can hold. The breach is therefore rectangular, vertical-sided and essentially instantaneous, and its width is a whole number of monoliths. This is the failure mode assumed here, and it decides how the breach is sized.

HEC-RAS offers five methods to compute breach parameters: MacDonald and Langridge-Monopolis (1984), Froehlich (1995), Froehlich (2008), Von Thun and Gillette (1990) and Xu and Zhang (2009). All five are regression equations fitted to the recorded failures of earth and rockfill embankments, 42 to 182 cases each, in which water erodes a trapezoidal channel through the fill over half an hour to several hours. Their inputs are the volume and depth of water above the breach floor and, for Xu and Zhang, the erodibility of the fill and the type of embankment; the 'concrete-faced' type in that method is a concrete-faced rockfill dam, not a concrete gravity dam. None of the five was fitted to a concrete dam, and none can represent a block that slides out in minutes. Table 2 shows what they would give for the Majlas sunny-day pool (102.2 Mm³, 61.5 m of water above the breach floor): widths of the right order, because they scale with the reservoir, but formation times of one to three hours, ten to thirty times too long for a concrete dam. A breach that opens slowly gives a lower peak, so using them here would understate the wave. They were therefore not used.



Table 2 - The regression methods offered by the model, evaluated for the Majlas sunny-day pool, against the guidance adopted.

Method (HEC-RAS parameter calculator)	Fitted to	Width for Majlas (m)	Formation time (h)	Applies to a concrete gravity dam?
Froehlich (2008)	74 embankment failures	116 to 151	0.9	No: erosion of fill
Froehlich (1995)	63 embankment failures	of the same order	of the same order	No: erosion of fill
MacDonald and Langridge-Monopolis (1984)	42 embankment failures	from the eroded volume	2.6	No: erosion of fill
Von Thun and Gillette (1990)	57 embankment failures	209	0.9 to 1.5	No: erosion of fill
Xu and Zhang (2009)	182 earth and rockfill failures	needs the erodibility of the fill	needs the erodibility of the fill	No: 'concrete-faced' means a concrete-faced rockfill dam
Guidance for concrete gravity dams (below)	Concrete dam case histories	whole monoliths, up to 156 (half the crest)	0.05 to 0.3	Yes

For concrete gravity dams the United States federal guidance gives the breach as a number of whole monoliths with vertical sides, usually not more than half the crest length, formed in 0.1 to 0.5 hour (USACE), 0.1 to 0.3 hour (FERC) or 0.1 to 0.2 hour (NWS). Table 3 sets the three side by side.

Table 3 - Breach parameters for concrete gravity dams in the United States federal guidance.

Source	Breach width	Side slopes	Formation time (h)
USACE (1980, 2007), as tabulated in the HEC-RAS reference and in TD-39	Usually not more than 0.5 of the crest length, whole monoliths	Vertical	0.1 to 0.5
FERC Engineering Guidelines	Usually not more than 0.5 of the crest length, whole monoliths	Vertical	0.1 to 0.3
NWS (National Weather Service)	Usually not more than 0.5 of the crest length, whole monoliths	Vertical	0.1 to 0.2

The base case takes five monoliths, 85 m, the width of two spillway bays. The spillway blocks are the tallest of the dam and carry the largest load; a failure that takes out two bays removes the ogee and the piers between them, opens the full height of the dam down to the wadi bed and stays within the



federal ceiling of half the crest (156 m). The formation time is the short end of the guidance, 0.1 hour: a monolith that slides leaves the dam in minutes, and a shorter time gives the higher, more conservative peak. The weir coefficient of the opening is the upper value of the range HEC-RAS gives for a clean, sharp-edged breach in a gravity dam (1.44 to 1.66 in SI units). Each of the three is then varied in the sensitivity runs to the bounds in Table 4: one bay and half the crest for the width, twice and half the formation time, and the lower coefficient.

Table 4 - Breach parameters of the base case and the bounds tested.

Parameter	Base case	Sensitivity bounds	Basis
Failure mode	Sliding of whole monoliths, vertical sides	-	Concrete gravity dam
Breach width	85 m, five monoliths (two spillway bays)	51 m (three monoliths, one bay) and 153 m (nine monoliths, about half the crest); 119 m on the flood day	Federal guidance: whole monoliths, at most half the crest
Breach bottom	23.0 m a.s.l.	-	Wadi bed at the axis: the block leaves down to its foundation contact
Formation time	0.1 h (6 minutes)	0.05 h (3 minutes) and 0.3 h (18 minutes)	Short end of USACE 0.1 to 0.5 h, FERC 0.1 to 0.3 h, NWS 0.1 to 0.2 h
Weir coefficient of the opening	1.66 (SI)	1.44	HEC-RAS range for a gravity-dam breach, 2.6 to 3.0 in US units
Opening	Sine-wave progression to the full width	-	A fast, smooth opening; the model option for a concrete dam



Two hand calculations bound the peak outflow before any model result is trusted. The weir equation applied to the fully open breach at the starting pool gives the upper value:

$$Q_p = CB(H - Z_b)^{1.5} \quad (1)$$

where:

Q_p : peak outflow through the breach (m^3/s)

C: weir coefficient of the opening, $1.66 \text{ m}^{0.5}/\text{s}$ in the base case

B: breach width, 85 m in the base case

H: pool level at failure (m a.s.l.)

Z_b : breach bottom, 23.0 m a.s.l.

The Ritter solution for the sudden removal of a wall onto a dry bed gives the lower value, because it accounts for the drop of the water surface at the opening:

$$Q_p = \frac{8}{27} B \sqrt{g} (H - Z_b)^{1.5} \quad (2)$$

where:

g: gravity, 9.81 m/s^2

For the base breach and the full supply level the weir equation gives about $68,000 \text{ m}^3/\text{s}$ and the Ritter solution about $38,000 \text{ m}^3/\text{s}$. The modelled sunny-day peak of $45,312 \text{ m}^3/\text{s}$ lies between them, lower than the weir value because the pool drops during the six minutes of formation and the tailwater rises in the stilling basin.

2.3. SCENARIOS AND THE REFERENCE CASE

Three scenarios frame the question. Each starts with the reservoir at the full supply level:

- **Sunny-day failure (S1).** No flood in the wadi, the reservoir at the full supply level, the dam fails without warning. This gives the sharpest wave and the least warning, because nobody expects a flood; it is the case the emergency plan is written for.



- **Flood-day failure (S2).** The probable maximum flood is passing and the dam fails at the moment the reservoir reaches its highest level, when the load on the dam is greatest. That level and moment are read from the no-failure run of the same flood.
- **Baseline, the PMF without failure (S3).** The same flood with the dam holding. It is the reference: only what the failure adds to it counts against the dam, and it also confirms the spillway capacity.

The three are summarised with their default conditions in Figure 3, and their names and settings are listed in chapter 3.



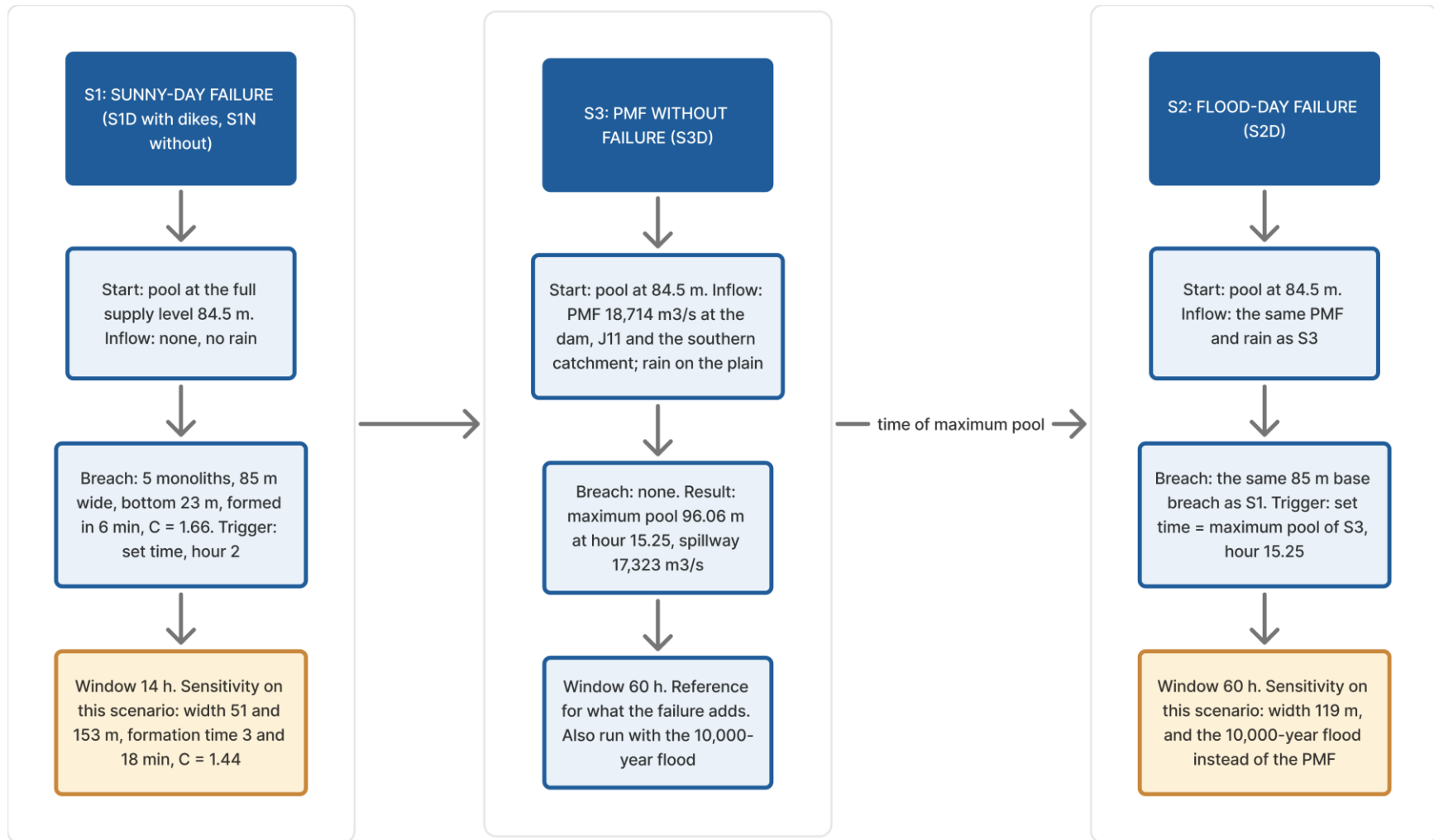


Figure 3 - The three scenarios and their default conditions.

For every consequence measure, the dam's contribution is the difference between the failure run and its no-failure twin:

$$\Delta C = C_{failure} - C_{no\ failure} \quad (3)$$

where:

ΔC : the dam's contribution to any consequence measure C: depth, area, people, hazard class

$C_{failure}$, $C_{no\ failure}$: the measure in the failure run and in its no-failure twin

2.4. HYDRAULIC MODEL

The flood is routed in two dimensions over one mesh from the reservoir to the sea, with the dam as a weir connection whose profile carries the spillway and the non-overflow crest and into which the breach is cut. The inflows to the reservoir and from the two side catchments, and the rain on the plain, come from the hydrology study. Figure 4 shows the order in which the model was assembled, from the terrain to the checks made before a result was used; chapter 4 gives every input as built.



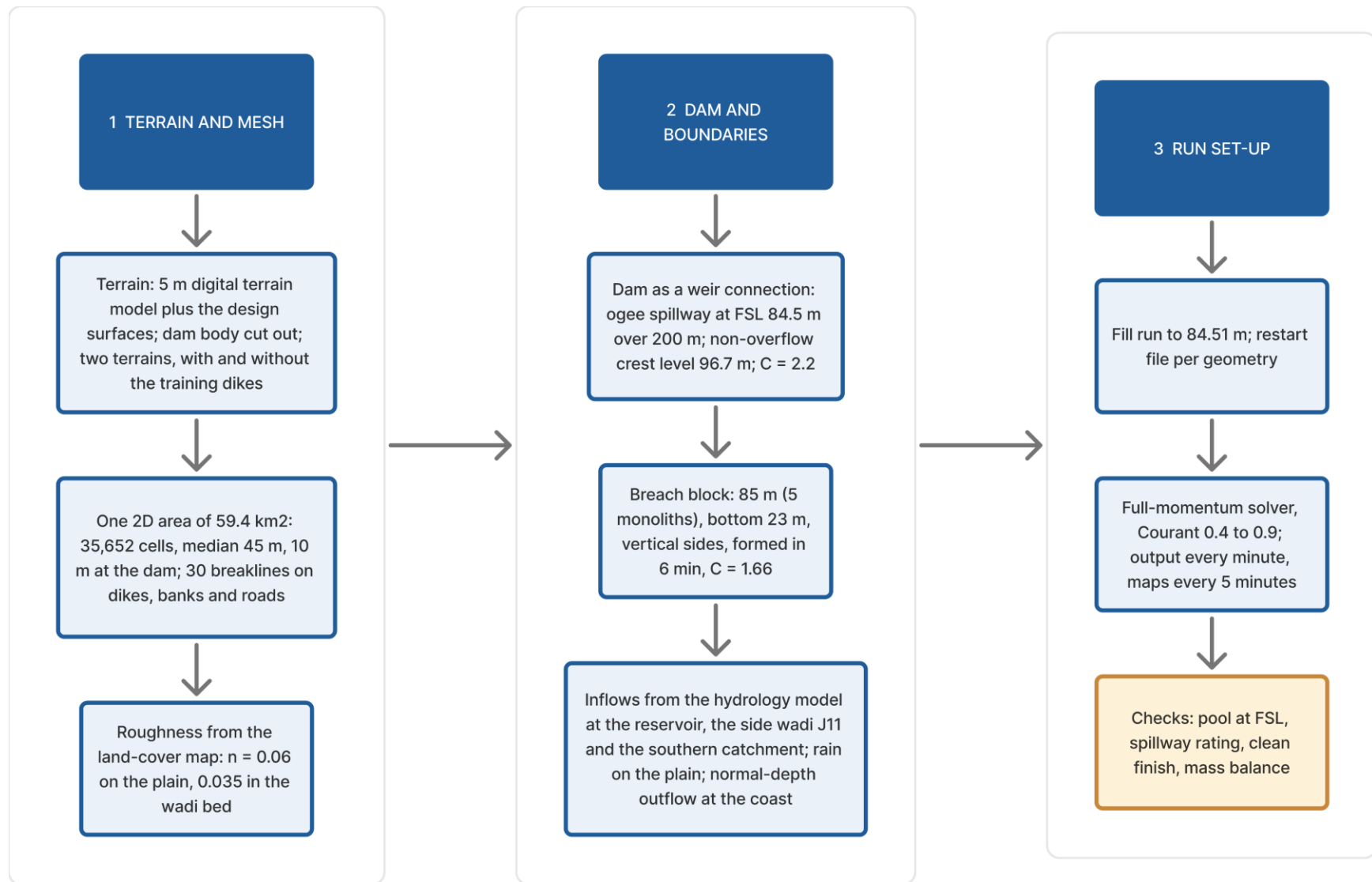


Figure 4 - How the hydraulic model was assembled.



Two relations govern the model at the dam and in the solver: the spillway follows the ogee equation, and the time step is kept below the Courant limit:

$$Q = C_s L H^{1.5} \quad (4)$$

where:

Q: spillway discharge (m³/s)

C_s: discharge coefficient of the ogee, 2.2

L: net crest length, 200 m

H: head above the crest at 84.5 m (m)

$$Cr = \frac{V \Delta t}{\Delta x} \quad (5)$$

where:

Cr: Courant number, kept between 0.4 and 0.9 by halving and doubling the step

V: flow velocity plus wave celerity (m/s)

Δt: time step (s)

Δx: cell size in the direction of flow (m)

2.5. CONSEQUENCES

For each run the model gives, on a 2 m grid, the maximum depth and velocity, their product, the time at which the water first arrives and the duration of flooding. Depth and velocity are combined into the



six Australian (AIDR) flood hazard classes, from H1, safe for people and vehicles, to H6, where buildings fail; a cell takes the highest class that any of the three limits gives:

$$D \times V = \text{depth} \times \text{velocity} \quad (6)$$

where:

D: maximum depth (m)

V: maximum velocity (m/s)

D × V: their product (m²/s), the measure of the force on a person or a wall

Table 5 - AIDR flood hazard classes: a cell is in the class whose limits it exceeds.

Class	D × V limit (m ² /s)	Depth limit (m)	Velocity limit (m/s)	Meaning
H1	0.3	0.3	2.0	Generally safe for people, vehicles and buildings
H2	0.6	0.5	2.0	Unsafe for small vehicles
H3	0.6	1.2	2.0	Unsafe for vehicles, children and the elderly
H4	1.0	2.0	2.0	Unsafe for people and vehicles
H5	4.0	4.0	4.0	Unsafe for people and vehicles; buildings vulnerable to structural damage
H6	above H5			Unsafe for everything; buildings fail

Figure 5 draws the same limits as curves in the depth-velocity plane; the classes are the same as those used in the flood-risk study of the downstream works, so the two studies can be read together.



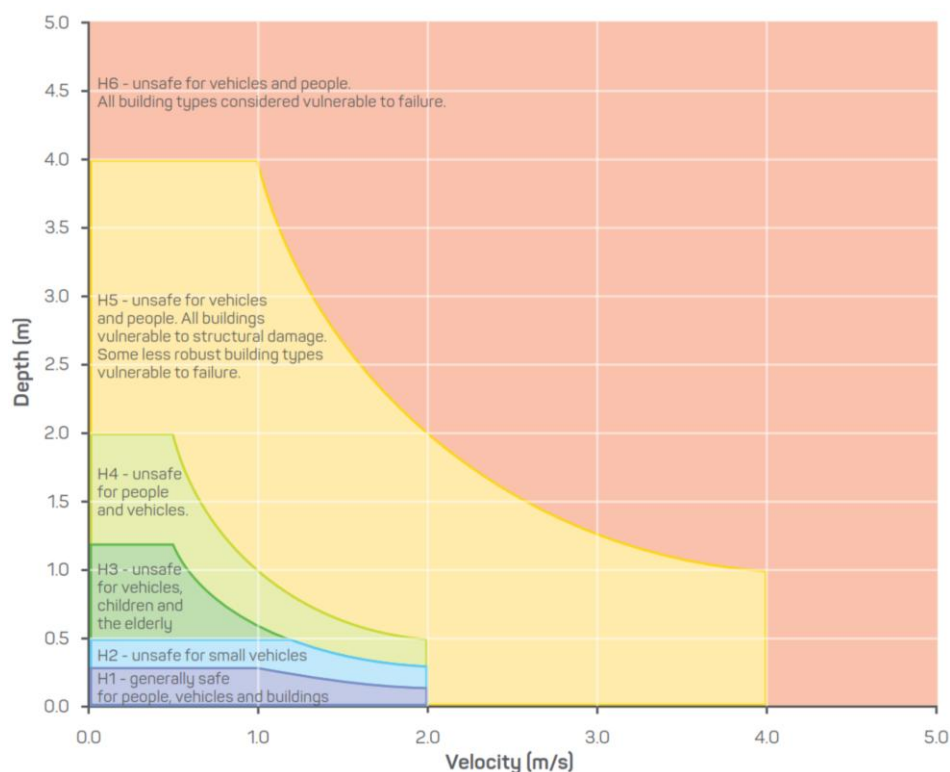


Figure 5 - General flood hazard vulnerability curves (Australian AIDR Guideline 7-3).

People are counted on a 100 m population grid¹ and properties on the plot layer of the flood-risk study, within the flooded area and per hazard class:

$$PAR = \sum P_i \text{ for every cell } i \text{ with } D_i > 0.3 \text{ m} \quad (7)$$

where:

PAR: population at risk

P_i : people in population cell i (100 m)

D_i : maximum depth at the centre of cell i (m)

¹ GHS-POP 2025, the global population grid of the European Commission's Joint Research Centre, at 100 m; the count is the sum of the cells whose centre lies in water deeper than 0.3 m.

Arrival is timed from each scenario's own trigger: a failure from the moment the breach opens, the flood without failure from the start of the storm. A place counts as reached when the water there rises 0.3 m above what was there before the trigger; for the flood-day failure, whose plain is already under the flood when the dam goes, it is the moment the failure makes the water 0.3 m deeper than the flood alone would at the same instant. People and plots are then counted by the time at which they are reached.

2.6. SENSITIVITY AND CHECKS

The breach width, the formation time, the weir coefficient of the opening and the flood itself are varied one at a time from the base case, so that each result carries a bound. Whether the training dikes needed their own set of runs was decided by the sunny-day pair, which isolates the dam's contribution:

$$\frac{|X_N - X_D|}{X_D} < 5\% \quad (8)$$

where:

X: flooded area, people in the flooded area and median arrival time

N, D: without and with the training dikes

Every run was checked for a clean solver finish and mass conservation before its products were used. Figure 6 shows the runs in the order they were made: the preparation of the model, the base runs with the dikes decision between them, and the sensitivity runs and products that followed.



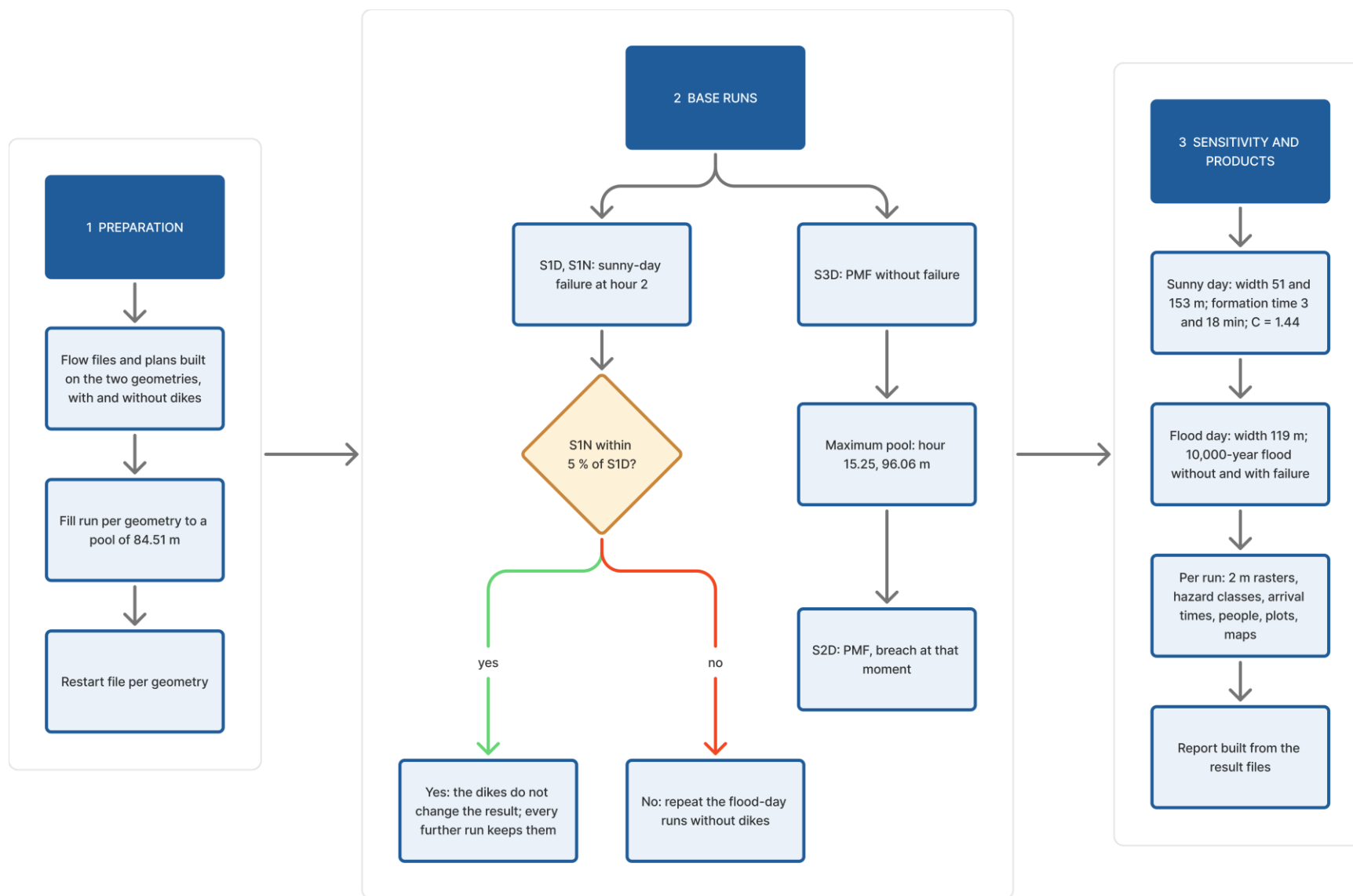


Figure 6 - The runs as executed, with the dikes decision.



2.7. STANDARDS AND GUIDANCE FOLLOWED

The analysis follows the international practice for concrete dams; Table 6 lists the documents and what each one was used for. The full references are at the end of the report.

Table 6 - Standards and guidance and what each was used for.

Document	Used for
USACE, HEC-RAS 6.6 User's, 2D Modeling and Hydraulic Reference Manuals (2024)	Model set-up, the weir connection and breach options, the breach weir coefficients, the solver
USACE, Using HEC-RAS for Dam Break Studies, TD-39 (Brunner, 2014)	Breach parameters by dam type, run set-up and checks for dam-break models
FERC, Engineering Guidelines, Chapter II (2015)	Breach width, side slopes and formation time for concrete gravity dams; inflow design flood
ICOLD Bulletin 111 (2020)	Scenarios, incremental reading of the consequences, presentation of the results
AIDR Guideline 7-3, Flood Hazard (2017)	The hazard classes H1 to H6 from depth and velocity



3. SCENARIOS AND RUNS

The scenarios of chapter 2 were run as listed in Table 7. The training dikes are in place in every run; the sunny-day failure was also run without them, to test whether they matter.

Table 7 - The scenarios.

Name	Scenario	Reservoir at the start	Inflow	Failure
S1D	Sunny-day failure, dikes in place	Full supply level, 84.5 m	None	Five monoliths at a set time, two hours into the run
S1N	Sunny-day failure, no dikes	Full supply level, 84.5 m	None	As S1D
S2D	Flood-day failure	Full supply level, 84.5 m	PMF, with rain on the plain	Five monoliths at the moment the reservoir reaches its maximum level, read from S3D
S3D	PMF without failure	Full supply level, 84.5 m	PMF, with rain on the plain	None

A sunny-day failure¹ gives the largest breach flow relative to the normal flow and the least warning. The flood-day failure is assumed to happen at the worst moment of the PMF, when the reservoir is at its highest and the load on the dam is greatest; this is the usual convention for a concrete gravity dam, whose critical case is the peak water load rather than overtopping. The PMF without failure is the reference that shows what the failure adds.

3.1. THE TRAINING DIKES

The training dikes were designed for the 200-year wadi flood. The two sunny-day runs, with and without them, isolate the dam's contribution with nothing else in the valley; Table 8 compares them.

¹ Sunny-day failure: a failure with no flood in the river, for example from a foundation problem or an earthquake. It gives the least warning, because nobody expects a flood.



Table 8 - Effect of the training dikes on the sunny-day failure.

Quantity	S1D, with dikes	S1N, without dikes	Difference
Peak flow at the dam (m ³ /s)	45,312	45,346	0.1%
Flooded area > 0.3 m (km ²)	17.0	17.1	0.4%
People in the flooded area	23,212	23,299	0.4%
Median arrival time on the plain	15 min	15 min	0.0%

A wave of this size overtops the dikes within minutes and spreads over the whole plain; the dikes neither contain it nor steer it to a measurable degree. The with-dikes runs therefore stand for both conditions, and every other scenario was run with the dikes in place.

3.2. SENSITIVITY RUNS

The sensitivity runs change one parameter at a time from the base case: the breach width (three, five, seven and nine monoliths), the formation time (3, 6 and 18 minutes), the weir coefficient of the breach opening, and the flood (the 10,000-year flood instead of the PMF). The breach mechanics are tested on the sunny-day scenario, where the dam's contribution stands alone; the flood day carries one width check and the smaller flood.

Table 9 - The sensitivity runs.

Name	What is changed from the base case	Purpose
S1D-W51, S1D-W153	Breach width 51 and 153 m (three and nine monoliths) instead of 85 m (five monoliths), sunny day	How much the peak and the flooded area depend on how many monoliths go
S1D-T3m, S1D-T18m	Formation time 3 min and 18 min instead of 6 min, sunny day	Whether the speed of the failure matters
S1D-C1.44	Breach weir coefficient 1.44 instead of 1.66, sunny day	Lower bound of the flow through the opening
S2D-W119	Breach width 119 m (seven monoliths), flood day	The width question checked on the flood day
S3D-F10000, S2D-F10000	10,000-year flood instead of the PMF, without and with failure	How the flood-day results change with a smaller flood



4. THE MODEL AS BUILT

4.1. EXTENT, TERRAIN, MESH AND ROUGHNESS

The model is one two-dimensional area of 59.4 km² from the upper reservoir to the shoreline, shown in Figure 7 with its boundary lines: the inflow to the reservoir, the two side catchments that join below the dam, and the two outflow lines at the coast. The terrain is the 5 m digital terrain model of the survey with the design surfaces of the dam and the dikes added; the dam body itself was cut out of it so that the dam connection alone controls the flow. Two terrains and two meshes were built, with and without the training dikes. Table 10 lists the mesh, the roughness and the solver settings.



The HEC-RAS model: 2D area, boundary lines and dam connection

Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

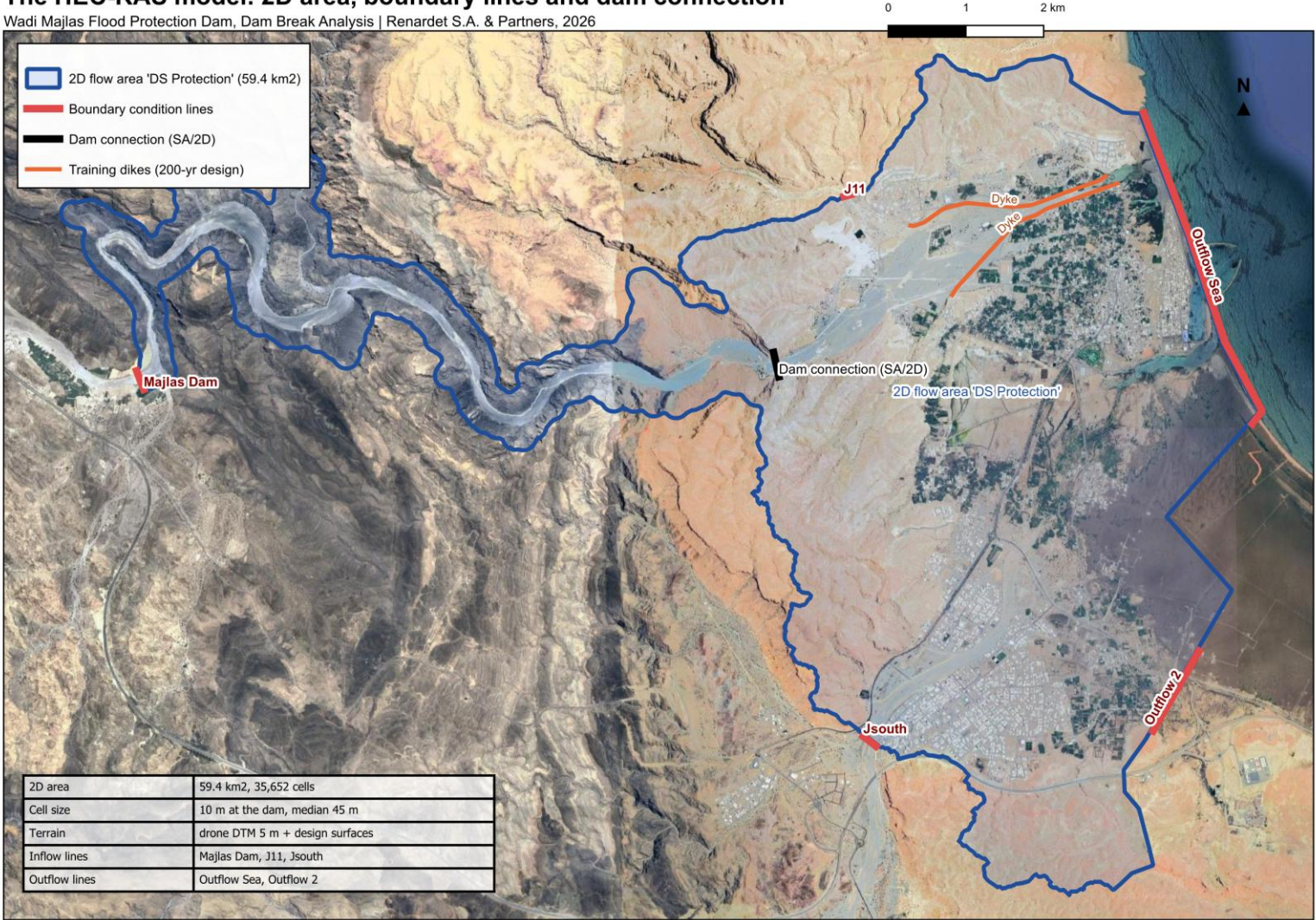


Figure 7 - The model: 2D flow area, boundary condition lines, dam connection and training dikes.



Table 10 - The 2D model as built.

Item	As built
Cells	35,652 with dikes, 35,878 without
Cell size	median 45 m, 10 % of cells below 21 m, largest 79 m; 10 m along the dam connection, 25 m beyond
Breaklines	30 lines at 20 to 25 m: dike crests, wadi banks, road embankments, dam axis
Roughness	$n = 0.06$ (plain, land cover default): 95.8 % of the area; $n = 0.035$ (wadi bed polygon): 4.2 % of the area
Solver	Full-momentum shallow-water equations ¹ , adaptive time step on a Courant number ² of 0.4 to 0.9, results every 1 minute, maps every 5 minutes
Simulation windows	14 h for the sunny-day runs (breach at hour 2), 60 h for the flood-day base runs, 36 h for the flood-day sensitivity runs

4.2. THE DAM AS DESIGNED

The model represents the dam as finalised in the design review of 20 September 2026. It is a roller-compacted concrete (RCC) gravity dam with a grout-enriched RCC facing on a straight axis, 312 m long at the crest and 72.5 m high above the wadi bed (110.5 m above the foundation at -15.0 m a.s.l.). The crest is 6.0 m wide at 95.5 m a.s.l. with a parapet to 96.7 m; the upstream face is 0.05H:1V and the downstream face 0.75H:1V. Figure 8 shows the upstream elevation: 17 monoliths separated by 16 contraction joints, B1 of 31.21 m against the left bank, B2 to B16 of 17.00 m each and B17 of 25.79 m against the right bank. The central blocks under the spillway stand on the foundation at -15.0 m; the abutment blocks step up the banks.

The spillway sits on the central blocks: a free ogee with its crest at 84.5 m a.s.l., the full supply level, in 5 bays of 40 m between 4 piers of 3 m (200 m net, 212 m gross, 218 m between the training walls), a stepped chute at 0.75H:1V and a USBR Type II stilling basin 55 m long with its floor at 15.5 m a.s.l. Its design capacity is 17,000 m³/s. The bottom outlet, one 2.8 m pipe in the left diversion culvert rated

¹ The Eulerian-Lagrangian scheme of the shallow-water equations (SWE-ELM), the full-momentum solver of HEC-RAS 6.6, which keeps the inertia of the wave that a diffusion-wave solver drops.

² Courant number: the distance the water moves in one time step divided by the cell size. Keeping it below one keeps the solution stable.



120 m³/s, is closed in every run. Figure 9 and Figure 10 show the non-overflow and spillway sections with their levels; Table 11 lists the values and where each one enters the model.

Table 11 - The dam as designed and where each value enters the model (design review, 20 September 2026).

Item	As designed	In the model
Dam type	Roller-compacted concrete (RCC) gravity dam with grout-enriched RCC facing	Failure by sliding of whole monoliths, vertical breach sides (chapter 2)
Crest / parapet top	95.5 / 96.7 m a.s.l.	Non-overflow crest of the connection at 96.7 m: the parapet is credited, so the PMF pool of 96.06 m does not overtop the dam
Crest width / length	6.0 m / 312 m	Weir width 6 m; connection line along the axis
Height above wadi bed / foundation	72.5 m / 110.5 m	Wadi bed at the axis 23.0 m a.s.l. = breach bottom
Foundation level	-15.0 m a.s.l.	Below the breach bottom; not modelled
Faces, upstream / downstream	0.05H:1V / 0.75H:1V	Vertical breach sides; the chute slope is in the terrain downstream
Monoliths	17 blocks: 31.21 m, 15 × 17.00 m, 25.79 m; 16 joints	Breach width in whole 17 m blocks: 51, 85 (base), 119 and 153 m = 3, 5, 7 and 9 blocks
Spillway	Free ogee at 84.5 m, 5 × 40 m bays, 4 × 3 m piers, net 200 m, 17,000 m ³ /s	Weir profile at 84.5 m over 200 m of the connection; ogee coefficient 2.2
Stilling basin	USB Type II, 55 m, floor 15.5 m a.s.l.	In the terrain
Bottom outlet	One 2.8 m pipe, intakes 36 and 44 m, 120 m ³ /s	Closed
Reservoir	102.2 Mm ³ at 84.5 m; 150.7 Mm ³ at 96.0 m	Storage from the terrain: 101.7 Mm ³ at 84.51 m after the fill





Figure 8 - Upstream elevation of the dam: the 17 monoliths B1 to B17 with their widths, crest 95.5 m, parapet 96.7 m and the foundation at -15.0 m (design review, September 2026).

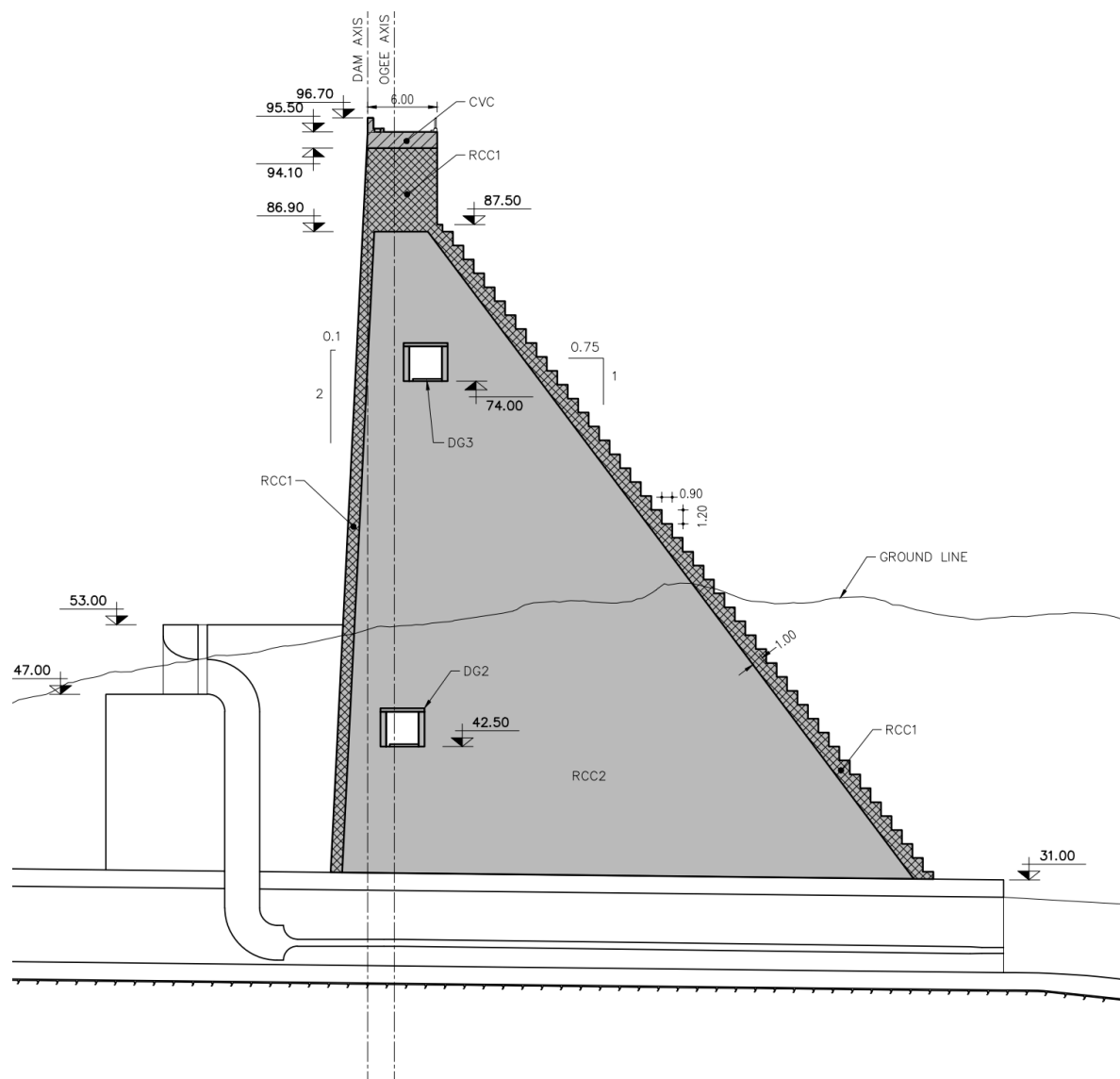


Figure 9 - Section through a non-overflow block: crest 95.5 m, parapet 96.7 m, galleries at 74.0 and 42.5 m, faces 0.05H:1V and 0.75H:1V (design review, September 2026).

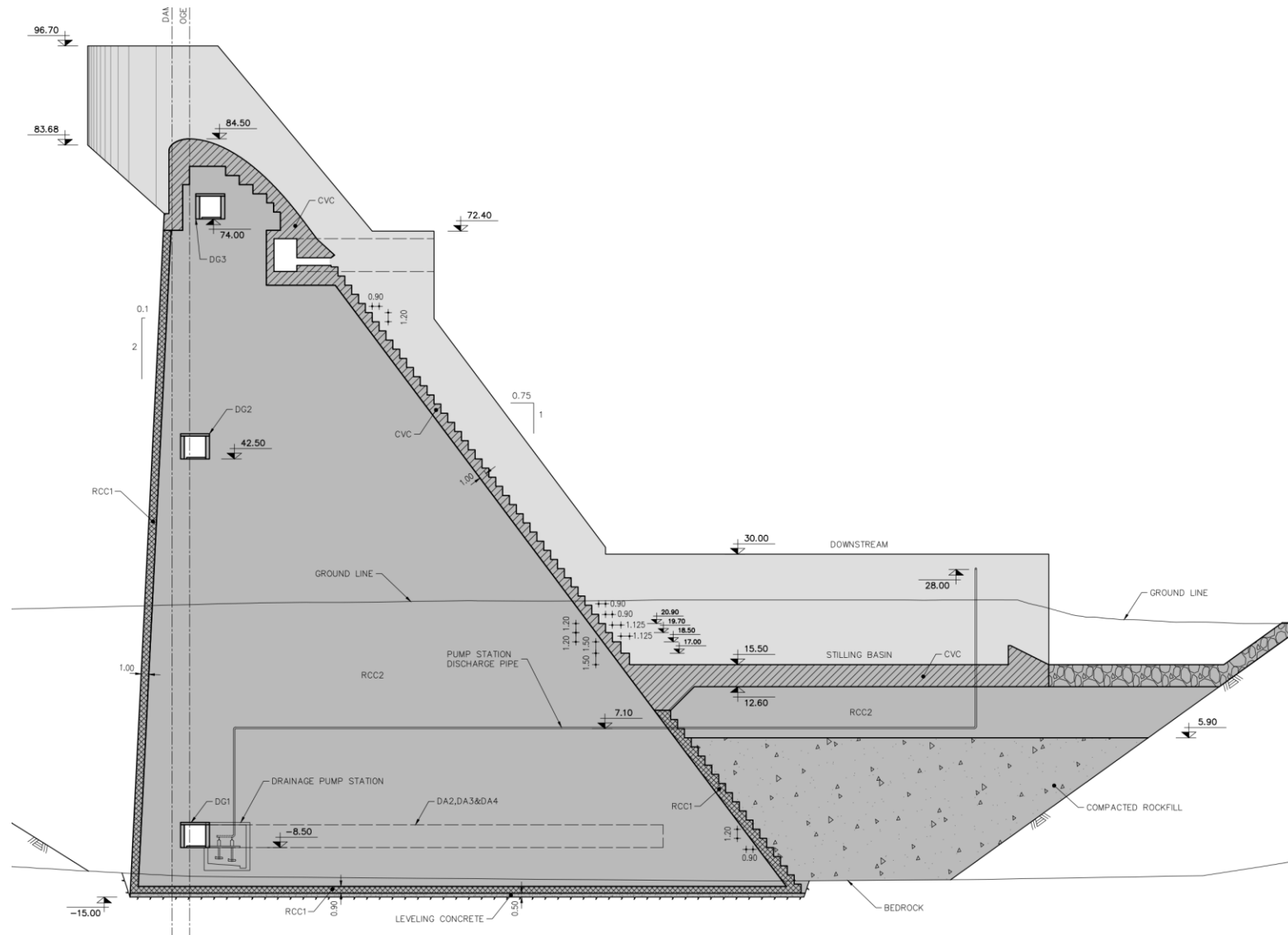


Figure 10 - Section through a spillway block: ogee crest 84.5 m, stepped chute, stilling basin floor 15.5 m, foundation -15.0 m (design review, September 2026).

4.3. THE DAM AND THE BREACH IN THE MODEL

The dam is a weir connection¹ whose profile carries the non-overflow crest at 96.7 m and the ogee at 84.5 m over 200 m of its length; weir coefficient 2.2, width 6 m, submergence limit 0.98. The breach is the base case of chapter 2, Table 12.

Table 12 - Breach parameters of the base case as entered in the model.

Parameter	Base value	Basis
Failure mode	Sliding of whole monoliths, vertical sides	Concrete gravity dam (chapter 2)
Breach width	85 m (five monoliths of 17 m)	Two spillway bays; within the guidance ceiling of half the crest
Breach bottom	23.0 m a.s.l.	Wadi bed at the axis
Formation time	0.1 h (6 min)	Short end of the guidance for concrete dams
Weir coefficient	1.66 (SI)	Upper value of the range for a gravity-dam breach, 1.44 to 1.66
Progression	Sine wave	A fast, smooth opening
Trigger, sunny day	Set time, 2 h into the run	Gives the model two hours to settle on the restart
Trigger, flood day	Set time = moment of maximum pool in S3D	Failure at the highest load

4.4. BOUNDARY CONDITIONS

Three hydrographs feed the model: the inflow to the Majlas dam reservoir, the side wadi J11 and the southern catchment of 159 km², both of which join the wadi below the dam. They come from the hydrology study's catchment model, extended with the southern catchment; Figure 11 shows that model and the three elements whose hydrographs are taken. Rain falls directly on the modelled area with the same storm, reduced by the areal factor of 0.76. The sunny-day runs have no inflow and no rain. Table 13 gives the design floods behind the two flood-day inputs, and Figure 12 the hydrographs and the rain as fed to the model.

¹ The model element that links two parts of the 2D area through a structure. Its profile carries the spillway crest and the non-overflow crest; the breach is cut into it.



HEC-HMS basin model 'Basin 1 Res2'

Red elements: hydrographs fed into HEC-RAS | Wadi Majlas Dam Break Analysis | Renardet S.A. & Partners, 2026

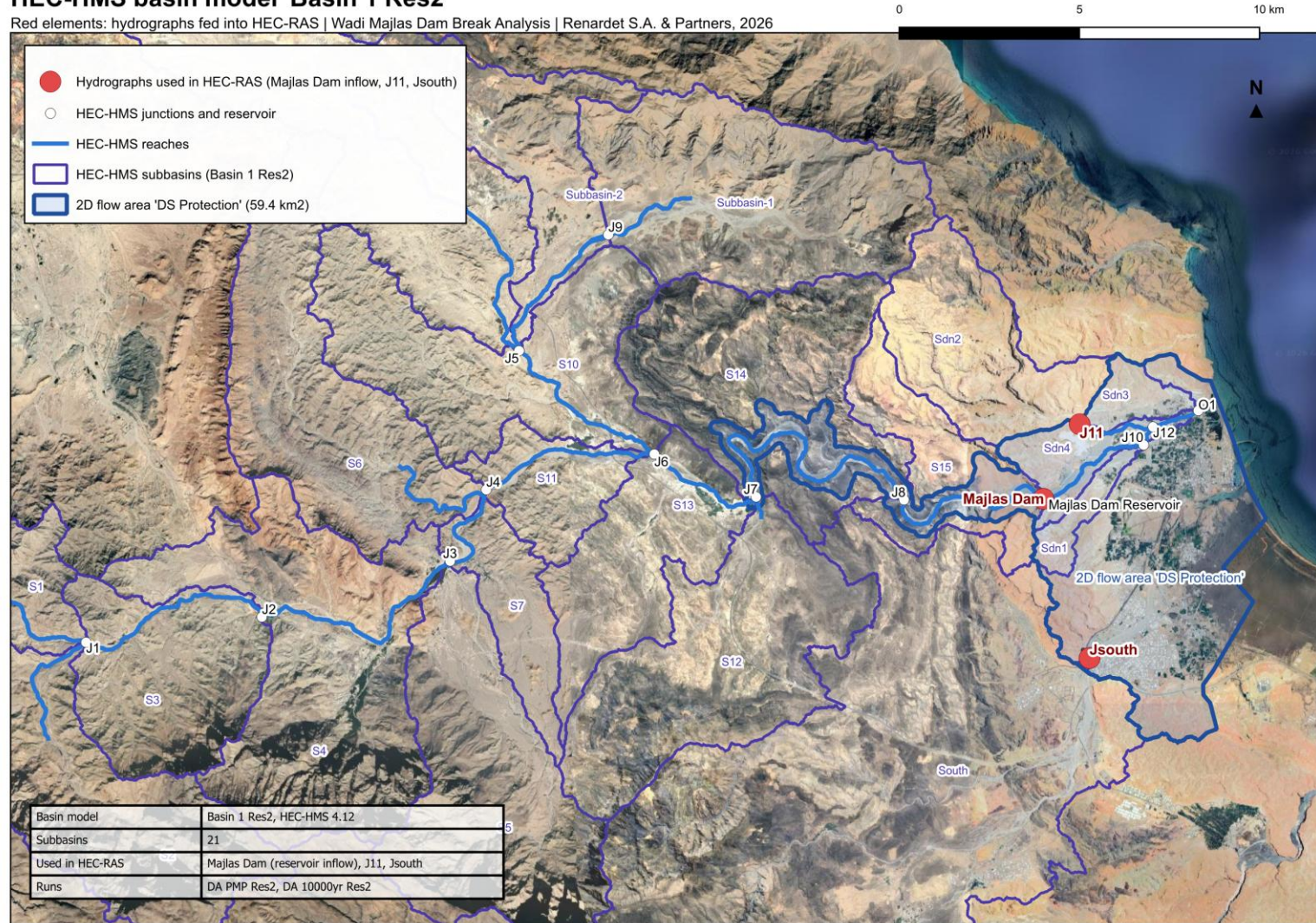


Figure 11 - The catchment model of the hydrology study and the three elements whose hydrographs feed the 2D model.

Table 13 - Design floods of the hydrology study used in this analysis (48-hour storm).

Flood	Storm rainfall (mm)	Peak inflow to the reservoir (m ³ /s)	Volume (Mm ³)
200-year (design flood of the training dikes)	251.0	4,563	90.5
10,000-year	685.1	14,156	278.0
Probable maximum flood (PMF)	1,306.6	18,714	549.4

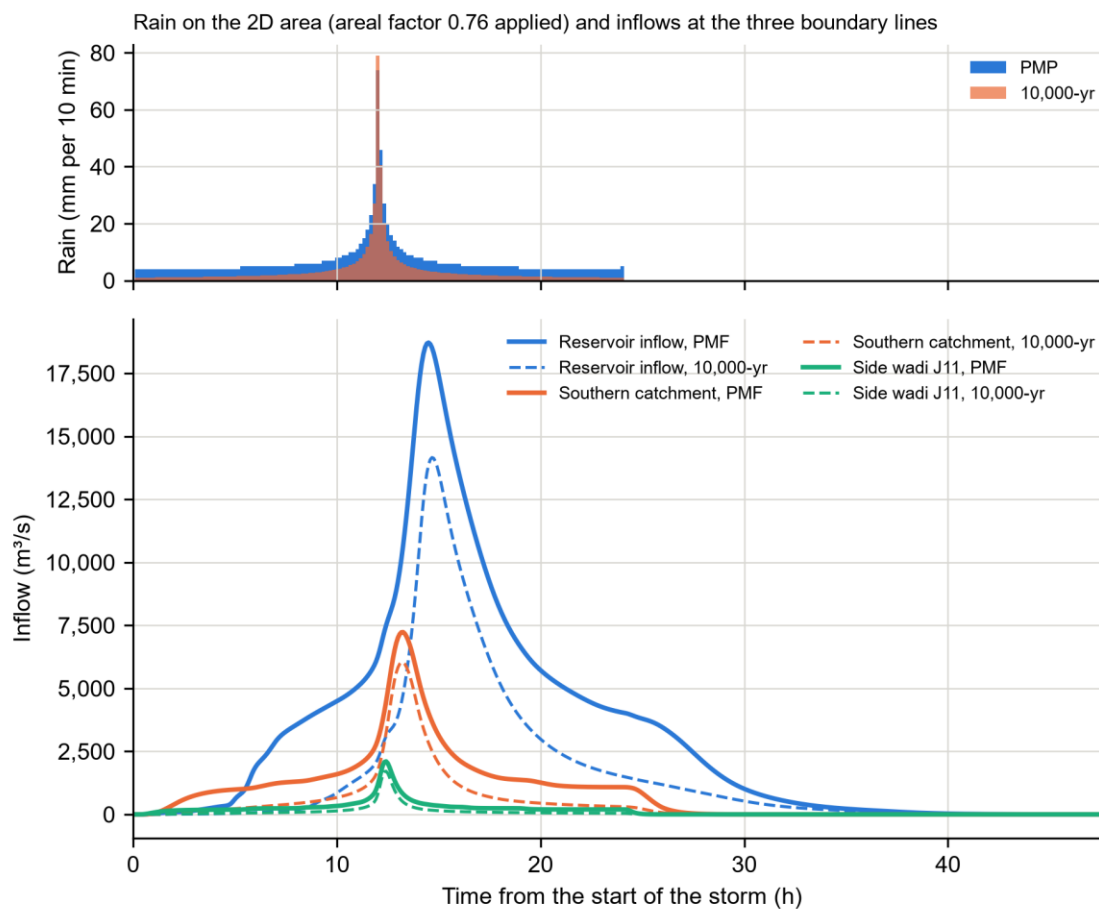


Figure 12 - The inputs to the flood-day runs: rain on the 2D area and the inflow hydrographs at the three boundary lines, PMF and 10,000-year flood.

Table 14 lists the boundary conditions of the flood-day runs; the coast is a free outflow at the sea level of the maritime tables¹.

Table 14 - Boundary conditions of the flood-day runs (times from the start of the storm).

Boundary	PMF	10,000-year flood
Inflow to the reservoir	18,714 m ³ /s at 14.5 h, 549.4 Mm ³	14,157 m ³ /s at 14.7 h, 278.0 Mm ³
Side wadi J11	2,101 m ³ /s at 12.4 h, 28.4 Mm ³	1,711 m ³ /s at 11.6 h, 12.8 Mm ³
Southern catchment	7,231 m ³ /s at 13.2 h, 156.9 Mm ³	6,034 m ³ /s at 12.4 h, 81.4 Mm ³
Rain on the 2D area	998 mm, peak 73.9 mm in 10 min	524 mm
Coast, two outflow lines	Free outflow at normal depth ²	same
Sea level	0.71 m a.s.l., mean higher high water	same

4.5. STARTING THE RUNS: THE FILL AND THE RESTART FILE

A run needs the reservoir full at the start. Instead of an initial-condition point, each geometry was filled once by a 48-hour run: 2,500 m³/s for 10.3 hours with one-hour ramps, 101.7 Mm³, which brings the pool to 84.51 m without spilling. The final state was saved as a restart file³, and every scenario starts from it with two hours of settling before a breach opens.

¹ Mean higher high water at Qurayyat, 2.6 m above chart datum, which lies 1.89 m below mean sea level: 0.71 m a.s.l. (Oman National Hydrographic Office, Maritime Book 2024). The shoreline terrain is higher, so the sea does not back water into the model; a storm surge coinciding with the flood is outside the scope of this study.

² Normal-depth outflow: the water leaves the model at the slope of the terrain, as it would along an open shore.

³ Restart file: a saved state of the model, water levels everywhere, used as the starting point of a run.



5. RESULTS: SUNNY-DAY FAILURE

The sunny-day runs answer the question the emergency plan asks first: how much water, how fast, and where, when the dam fails with no flood in the wadi. Table 15 gives the key results¹.

Table 15 - Sunny-day failure: key results.

Scenario	Description	Peak flow (m ³ /s)	Time of peak	Max pool (m)	Area (km ²)	Depth gorge / plain (m)	People
S1D	sunny-day failure, dikes in place	45,312	2 h 10 min	84.51	17.0	54 / 11.5	23,212
S1N	sunny-day failure, no dikes	45,346	2 h 10 min	84.51	17.1	54 / 11.3	23,299

The breach opens two hours into the run. Within six minutes the opening is 85 m wide down to the wadi bed, and the outflow peaks at 45,312 m³/s. The reservoir level falls from 84.51 m to below 40 m a.s.l. within one hour and 103 Mm³ leave the reservoir in about two hours. The wave fills the gorge to more than 40 m depth at the dam, reaches the coastal plain within 15 min of the breach, the residential centre within 15 min, the shoreline within 22 min and the last corner of the flooded area after 1 h 25 min. In the gorge below the dam the water stands 54 m deep; on the plain depths reach 11.5 m and velocities 12.7 m/s. Figure 13 and Figure 14 draw the flow past the dam and the reservoir level for the three scenarios; the sunny-day wave is the sharp one.

¹ Peak flow: the largest flow past the dam (breach and spillway together). Max pool: the highest reservoir level, in m a.s.l. Area: the plain and wadi covered by more than 0.3 m of water, reservoir excluded. Depth: the largest depth in the gorge below the dam and on the plain. People: residents of the flooded area.



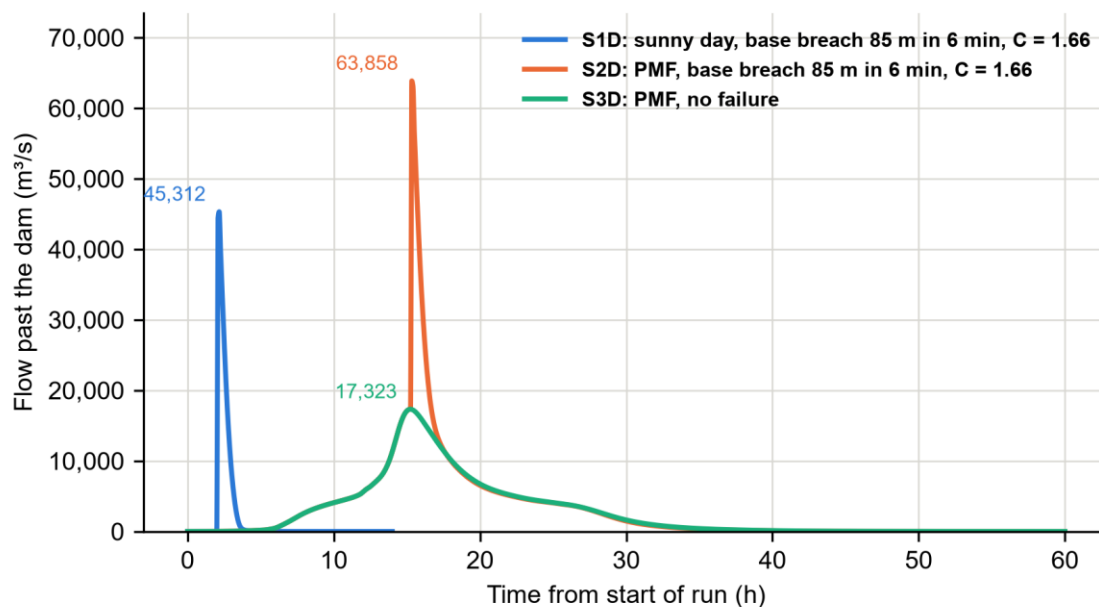


Figure 13 - Flow past the dam in the three scenarios.

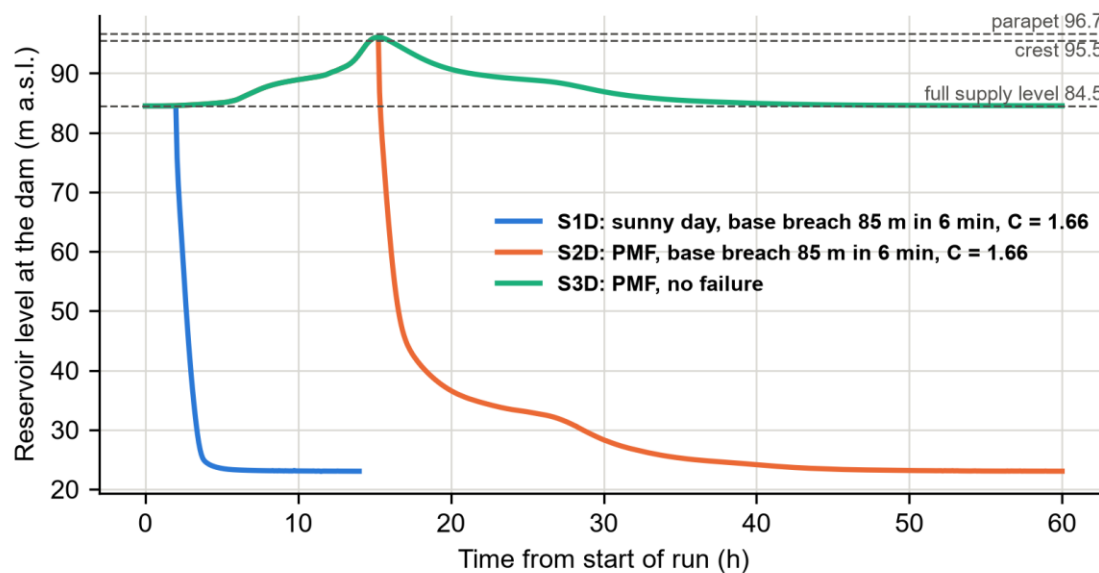


Figure 14 - Reservoir level at the dam in the three scenarios.

Figure 15 to Figure 18 map the maximum depth over the whole reach and, for Qurayat, the hazard class with the arrival of the wave drawn as isochrones, black for the earliest, greying with time. The isochrones are travel times of the water; the warning time available to people is shorter by the time it takes to detect the failure and to spread the alarm.

S1D: maximum depth

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

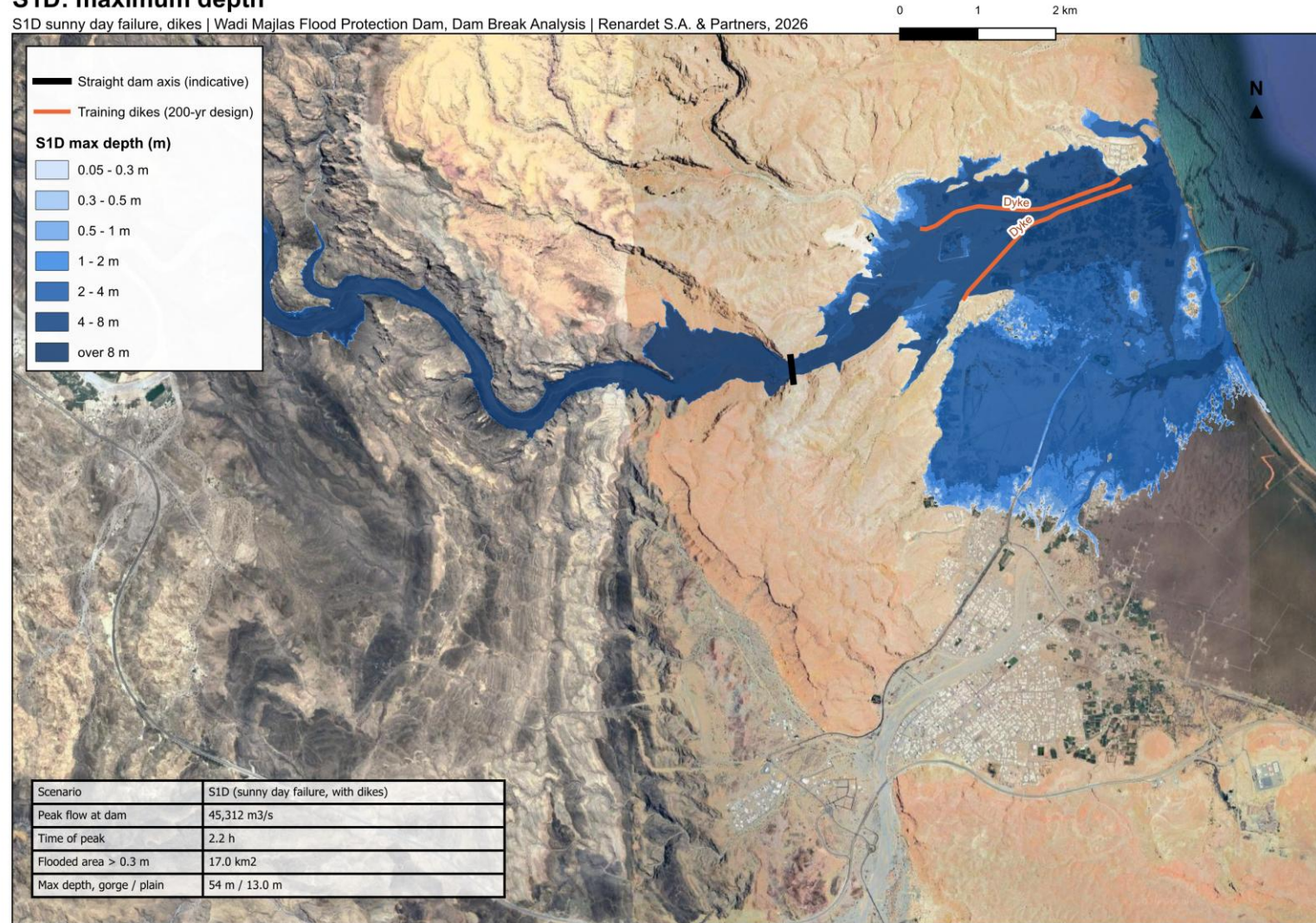


Figure 15 - S1D, sunny-day failure, dikes in place: maximum depth.

S1D: hazard class with arrival isochrones after the breach

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

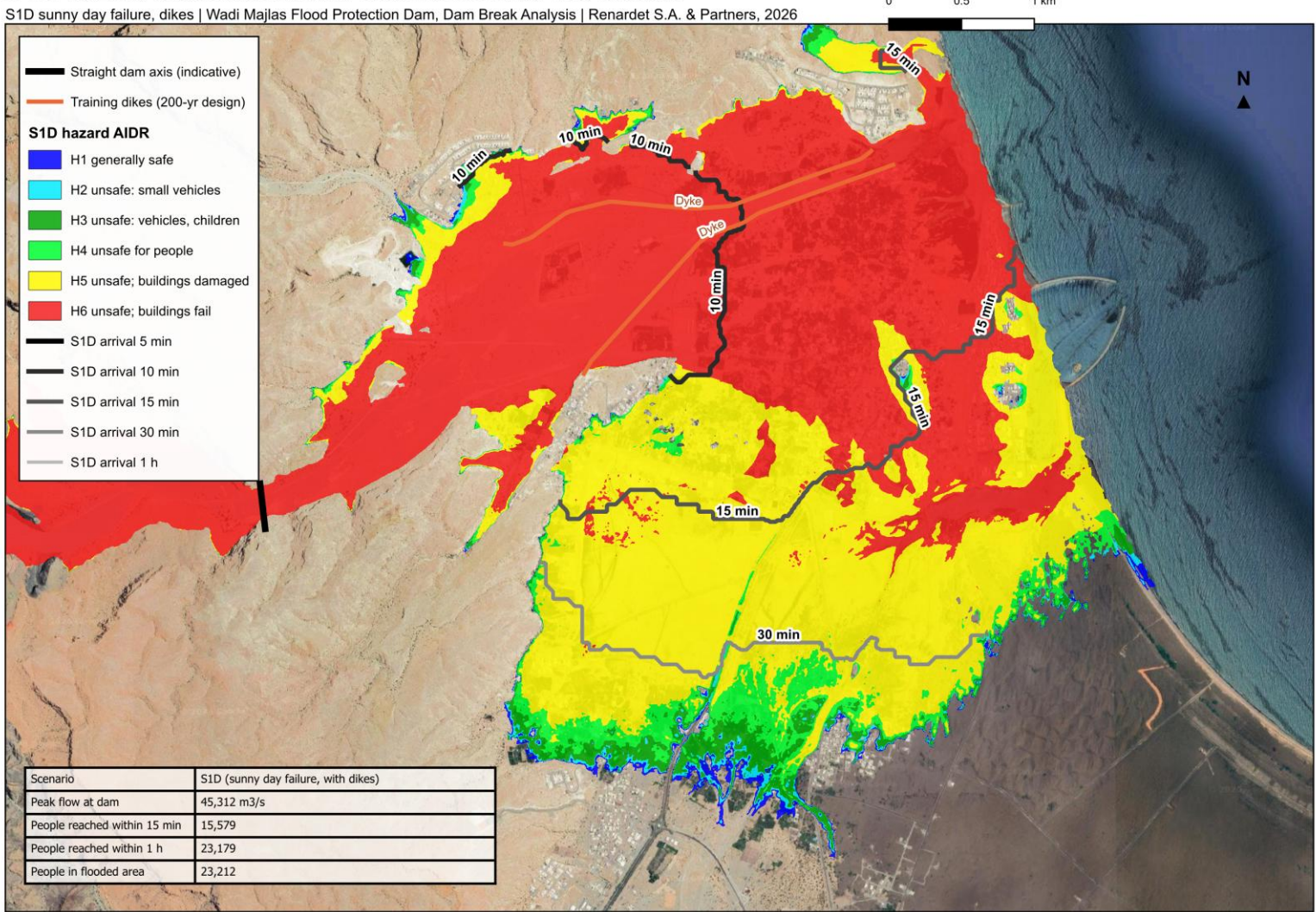


Figure 16 - S1D: flood hazard class (AIDR) with the arrival of a 0.3 m rise after the breach, isochrones black (earliest) to grey, Qurayat.



S1N: maximum depth

S1N sunny day failure, no dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

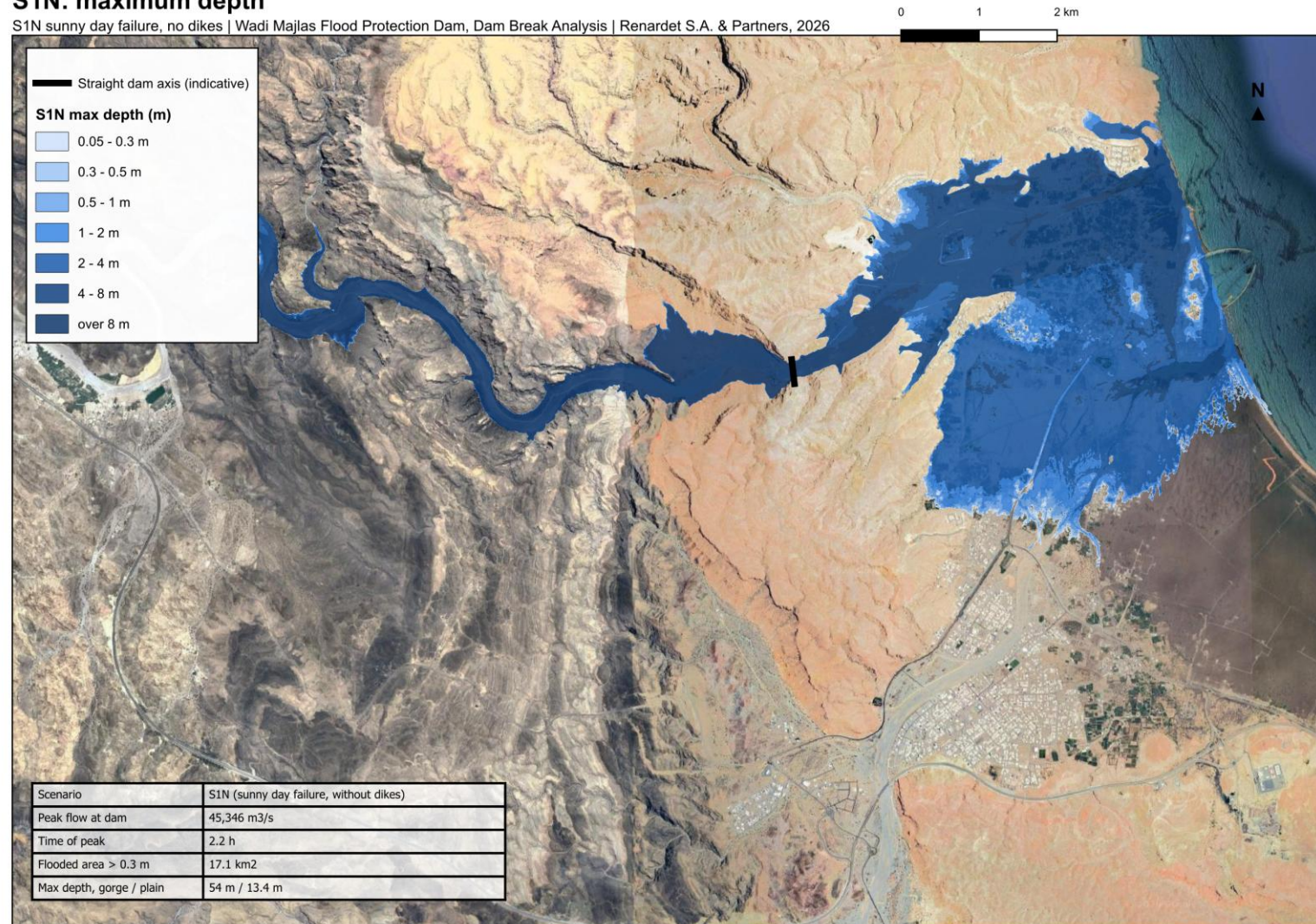


Figure 17 - S1N, sunny-day failure, no dikes: maximum depth.

S1N: hazard class with arrival isochrones after the breach

S1N sunny day failure, no dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

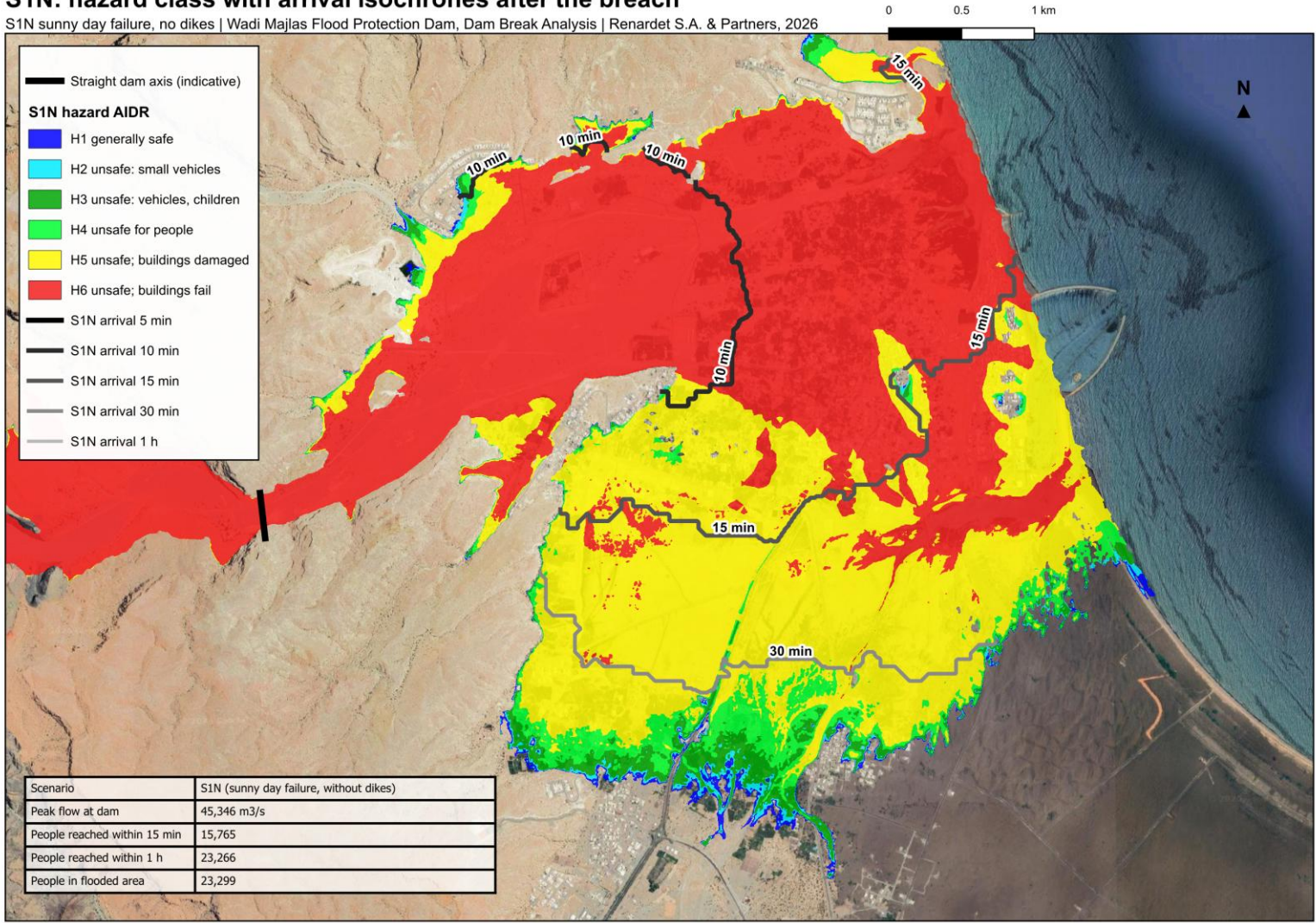


Figure 18 - S1N: flood hazard class (AIDR) with the arrival of a 0.3 m rise after the breach, isochrones black (earliest) to grey, Qurayat.



6. RESULTS: FLOOD-DAY FAILURE

The flood-day runs add the failure to a valley already under the PMF; the difference to the no-failure run of chapter 7 is the dam's contribution. Table 16 gives the key results, for the PMF and for the 10,000-year flood.

Table 16 - Flood-day failure: key results.

Scenario	Description	Peak flow (m ³ /s)	Time of peak	Max pool (m)	Area (km ²)	Depth gorge / plain (m)	People
S2D	flood-day (PMF) failure, dikes in place	63,858	15 h 20 min	96.06	27.1	65 / 13.5	46,299
S2D-F10000	flood-day failure during the 10,000-year flood	58,752	15 h 40 min	93.75	26.4	62 / 13.1	43,376

The failure is triggered 15 h 15 min into the storm, when the reservoir stands at 96.06 m a.s.l., the maximum level of the no-failure run. The outflow peaks at 63,858 m³/s, against 17,323 m³/s over the spillway without failure. The flooded area grows only from 25.9 to 27.1 km², because the PMF has already covered the plain; what the failure adds is depth, force and speed. The area in the worst hazard class, H6, grows from 13.0 to 18.4 km², and the people in the flooded area from 44,088 to 46,299. The plain is under water before the dam fails, so the arrival that matters is that of the failure wave on top of the flood: the moment the failure makes the water 0.3 m deeper than the flood alone would. That wave reaches the residential centre 10 min after the breach and the shoreline 10 min after; chapter 9 counts the people it reaches.

During the 10,000-year flood the same failure, triggered at the 93.75 m pool, peaks at 58,752 m³/s against 12,386 m³/s without failure, floods 26.4 km² with 43,376 people, and adds 0.3 m to the water standing on 24,293 people within 15 minutes: the smaller flood lowers the peak by a tenth and changes little else.



S2D: maximum depth

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

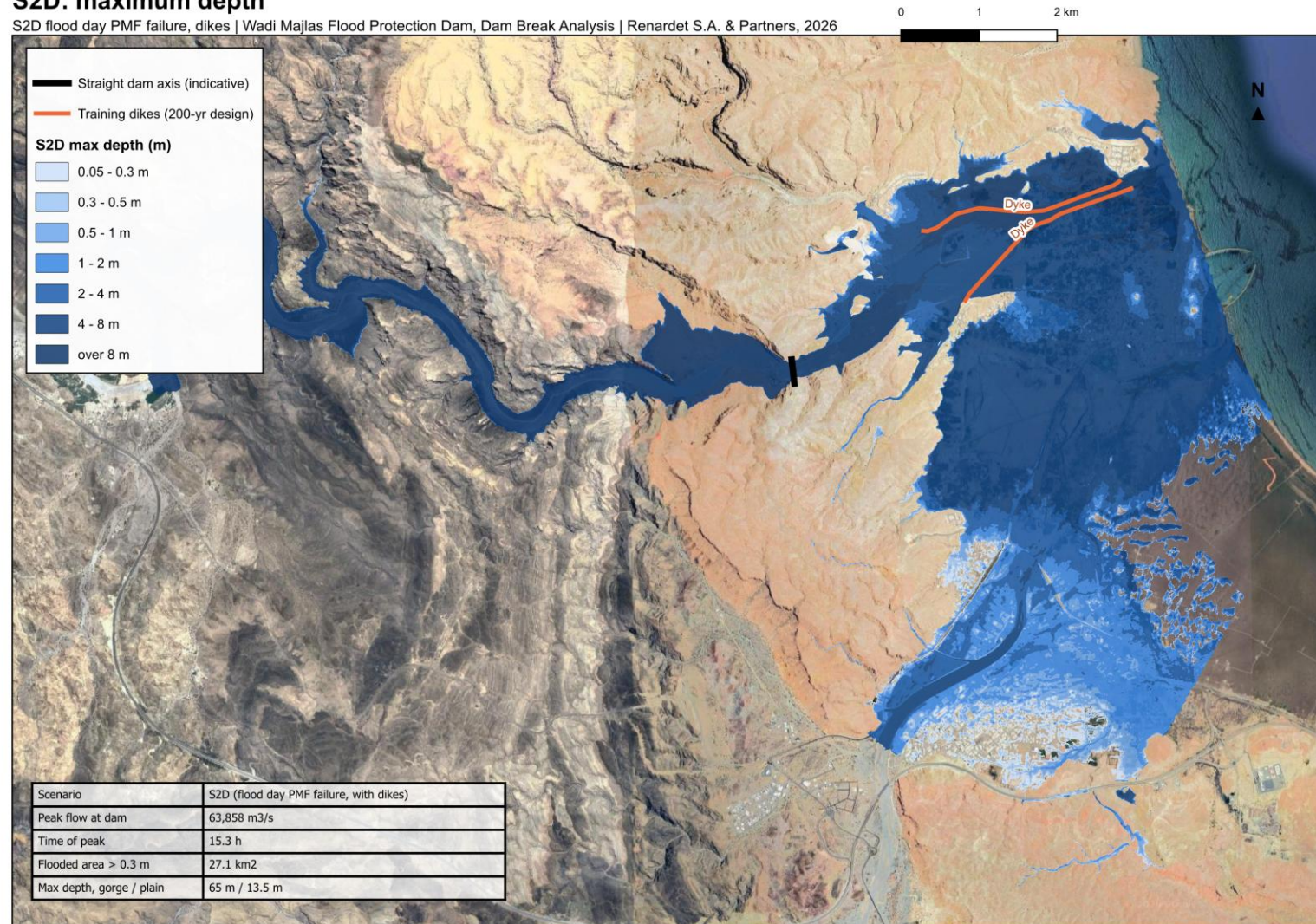


Figure 19 - S2D, flood-day (PMF) failure, dikes in place: maximum depth.

S2D: hazard class with arrival of the failure wave over the PMF

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

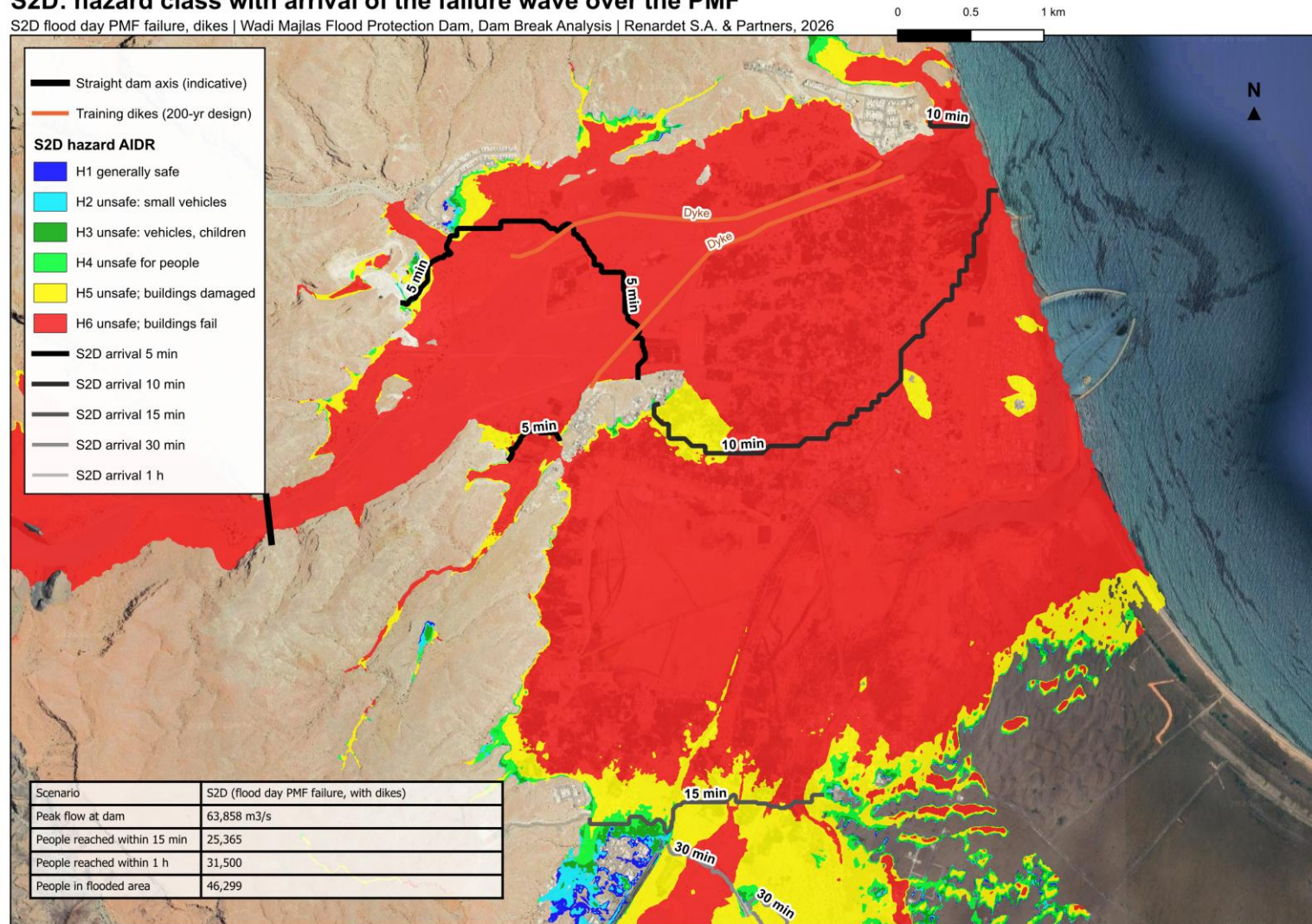


Figure 20 - S2D: flood hazard class (AIDR) with the arrival of the failure wave over the PMF (0.3 m extra depth), isochrones black (earliest) to grey, Qurayat.

7. RESULTS: THE PMF WITHOUT FAILURE

The no-failure runs show what the valley faces from the flood alone, and give the maximum reservoir level and the moment of the failure for the flood-day runs. Table 17 gives the key results for the PMF and for the 10,000-year flood.

Table 17 - PMF without failure: key results.

Scenario	Description	Peak flow (m ³ /s)	Time of peak	Max pool (m)	Area (km ²)	Depth gorge / plain (m)	People
S3D	PMF without failure, dikes in place	17,323	15 h 15 min	96.06	25.9	47 / 12.2	44,088
S3D-F10000	10,000-year flood without failure, dikes in place	12,386	15 h 30 min	93.75	24.8	46 / 12.0	39,083

The reservoir peaks at 96.06 m a.s.l., 0.56 m above the crest and 0.64 m below the parapet, and the spillway passes 17,323 m³/s against a design discharge of 17,000 m³/s: the dam is not overtopped. The flood covers 25.9 km² of the plain, much of it in the worst hazard class, but it builds up over hours: the residential centre is reached 1 h 06 min after the storm starts and the shoreline 1 h 38 min after.

The 10,000-year flood, run on the same model as a check, peaks at 93.75 m a.s.l., 1.75 m below the crest, with 12,386 m³/s over the spillway; it covers 24.8 km² of the plain with 39,083 people in it, and reaches the town 3 h 15 min after the storm starts, two hours later than the PMF.

Figure 21 maps the maximum depth, Figure 22 the hazard class with the arrival isochrones in hours, and Figure 23 the arrival time on its own, where the slow build-up of the flood over the plain is easier to read.



S3D: maximum depth

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

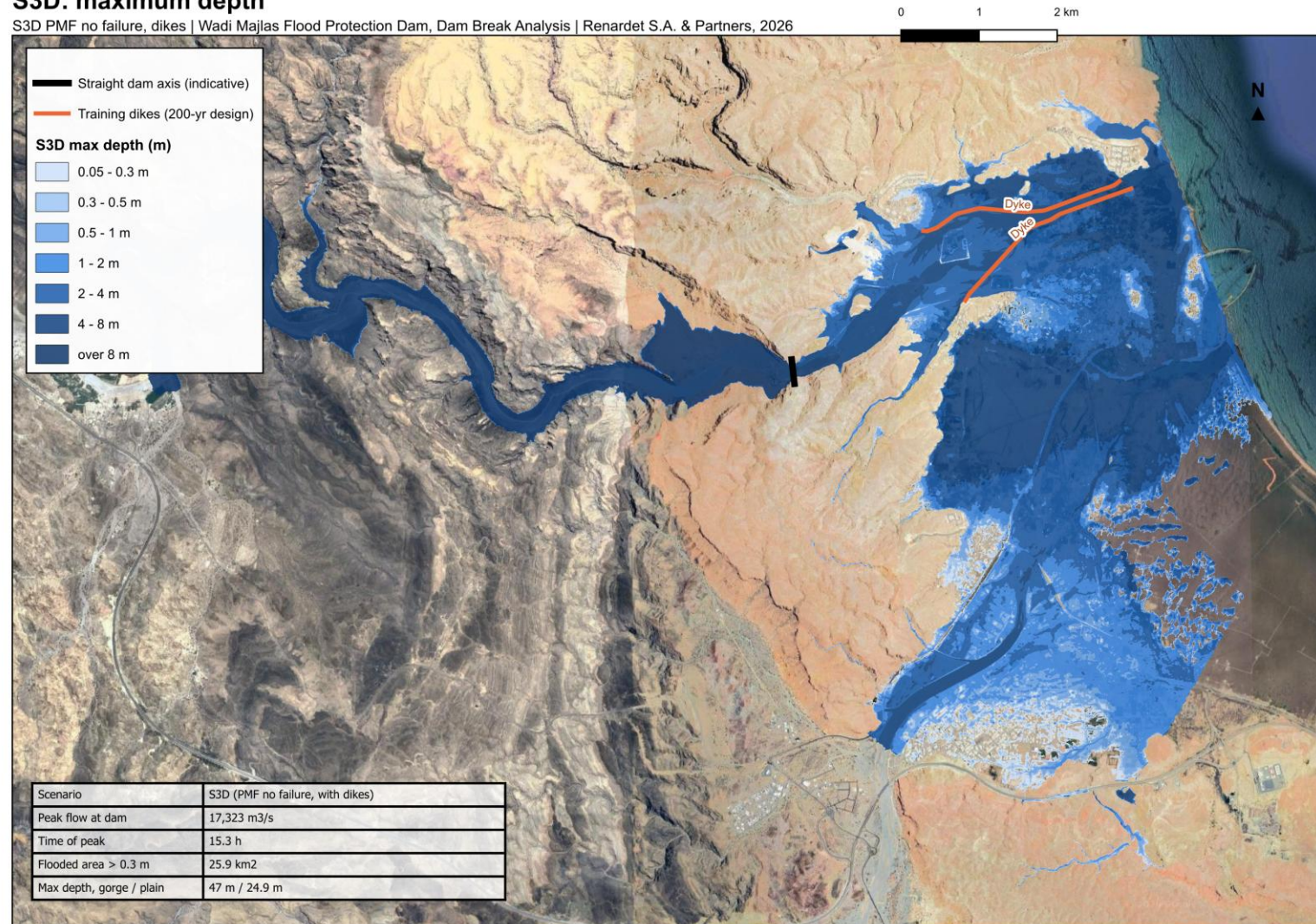


Figure 21 - S3D, PMF without failure, dikes in place: maximum depth.

S3D: hazard class with arrival isochrones after the start of the storm

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

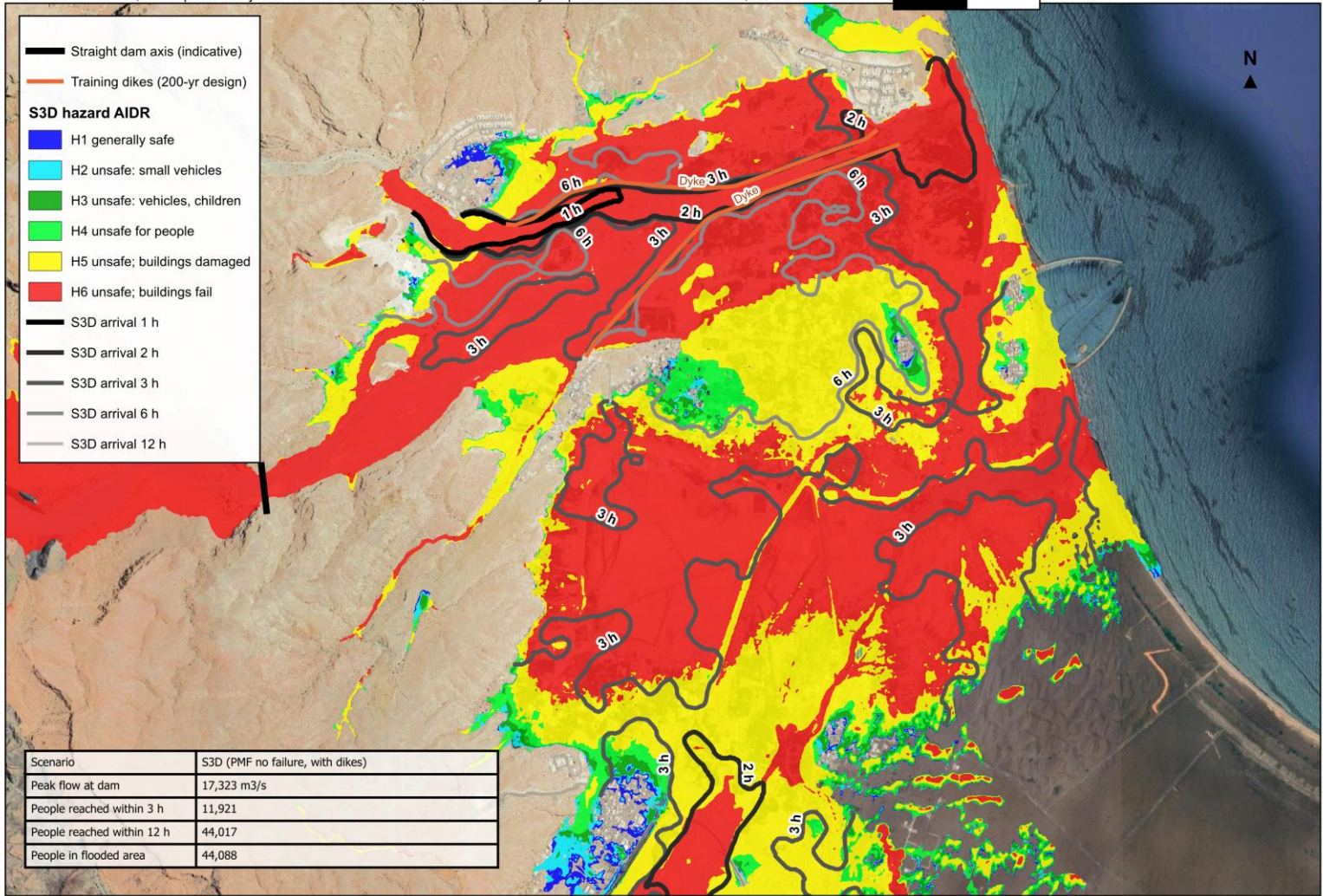


Figure 22 - S3D: flood hazard class (AIDR) with the arrival of a 0.3 m rise after the start of the storm, isochrones in hours, black (earliest) to grey, Qurayat.



S3D: arrival of a 0.3 m rise after the start of the storm

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

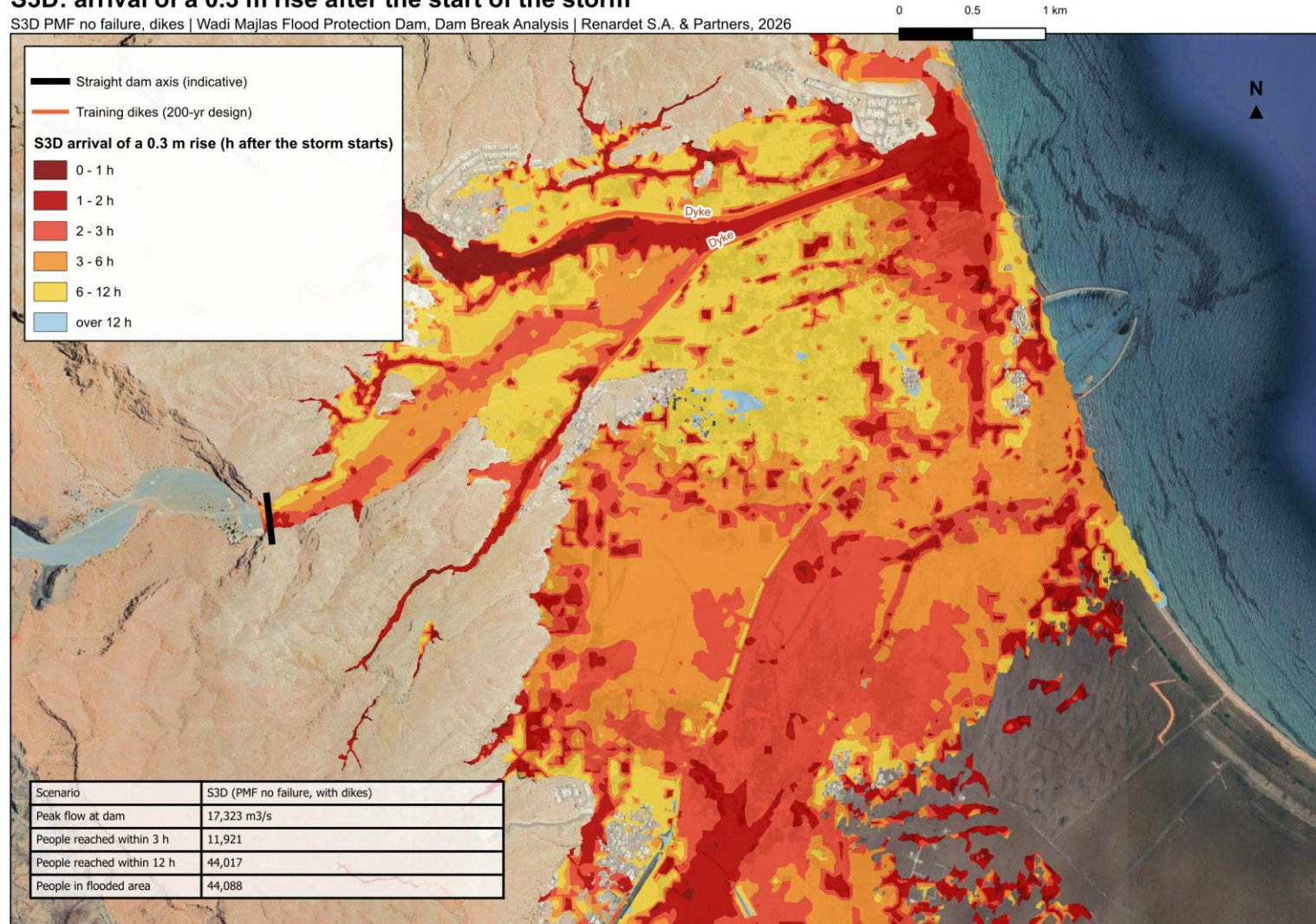


Figure 23 - S3D: arrival of a 0.3 m rise after the start of the storm, Qurayat.

8. SENSITIVITY OF THE RESULTS TO THE BREACH PARAMETERS

The breach is prescribed, not computed, so its width, speed and discharge coefficient were varied one at a time to see how far the results move; the flood was varied too. Table 18 gives the key results of the eight runs, with the base cases for comparison.

Table 18 - Sensitivity runs and base cases: key results.

Scenario	Description	Peak flow (m ³ /s)	Time of peak	Max pool (m)	Area (km ²)	Depth gorge / plain (m)	People
S1D	sunny-day failure, dikes in place	45,312	2 h 10 min	84.51	17.0	54 / 11.5	23,212
S1N	sunny-day failure, no dikes	45,346	2 h 10 min	84.51	17.1	54 / 11.3	23,299
S1D-W51	sunny-day failure, breach 51 m wide (three monoliths)	31,663	2 h 10 min	84.51	16.4	52 / 10.2	21,999
S1D-W153	sunny-day failure, breach 153 m wide (nine monoliths)	66,955	2 h 05 min	84.51	17.4	56 / 13.0	24,148
S1D-C1.44	sunny-day failure, breach weir coefficient 1.44	42,384	2 h 10 min	84.51	16.9	52 / 11.2	22,876
S1D-T3m	sunny-day failure, breach formed in 3 minutes	49,525	2 h 05 min	84.51	17.0	55 / 11.5	23,212
S1D-T18m	sunny-day failure, breach formed in 18 minutes	42,856	2 h 20 min	84.51	17.0	53 / 11.5	23,211
S2D	flood-day (PMF) failure, dikes in place	63,858	15 h 20 min	96.06	27.1	65 / 13.5	46,299
S2D-W119	flood-day failure, breach 119 m wide (seven monoliths)	79,030	15 h 20 min	96.06	27.5	66 / 21.3	46,960
S2D-F10000	flood-day failure during the 10,000-year flood	58,752	15 h 40 min	93.75	26.4	62 / 13.1	43,376
S3D	PMF without failure, dikes in place	17,323	15 h 15 min	96.06	25.9	47 / 12.2	44,088
S3D-F10000	10,000-year flood without failure, dikes in place	12,386	15 h 30 min	93.75	24.8	46 / 12.0	39,083

Figure 24 to Figure 28 draw the flow past the dam for each family of runs, and Figure 29 the peak flow and the flooded area of every run.



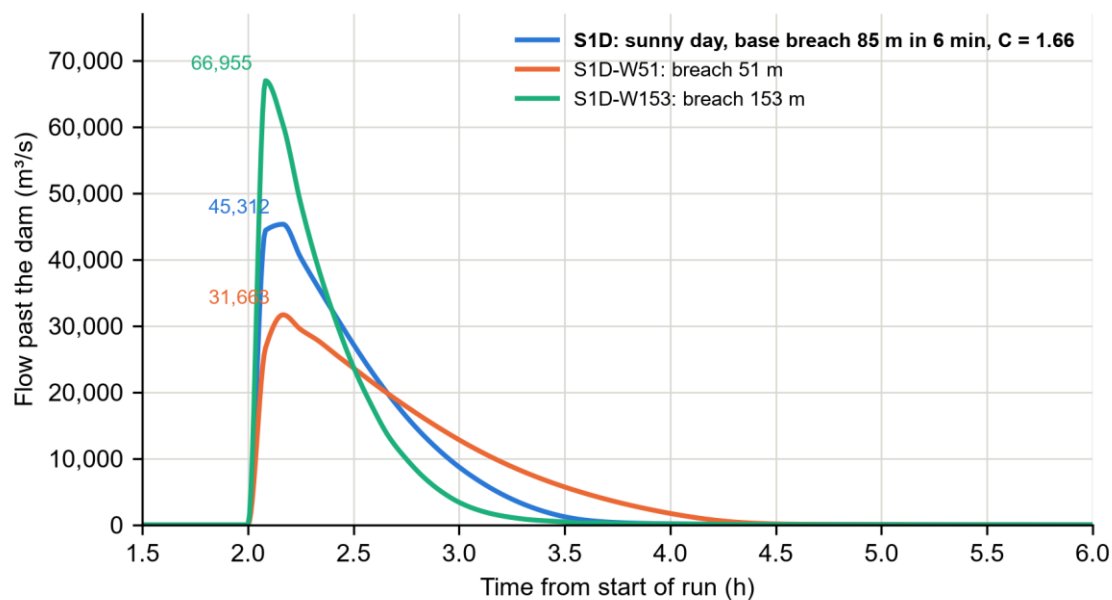


Figure 24 - Flow past the dam for the sunny-day breach-width runs (51, 85 and 153 m).

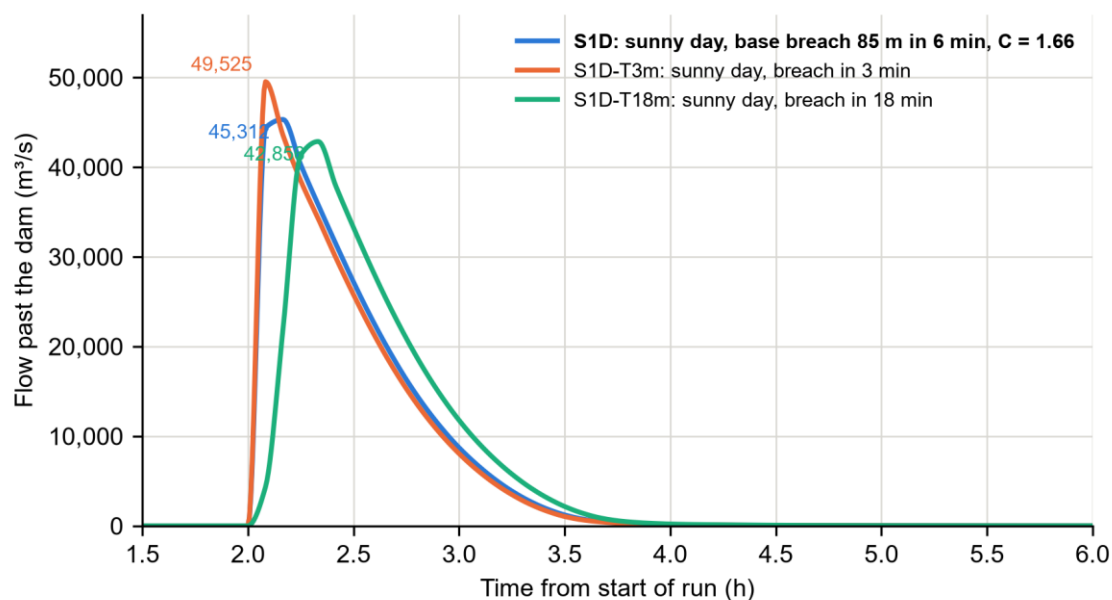


Figure 25 - Flow past the dam for the three formation times, sunny day (3, 6 and 18 min).

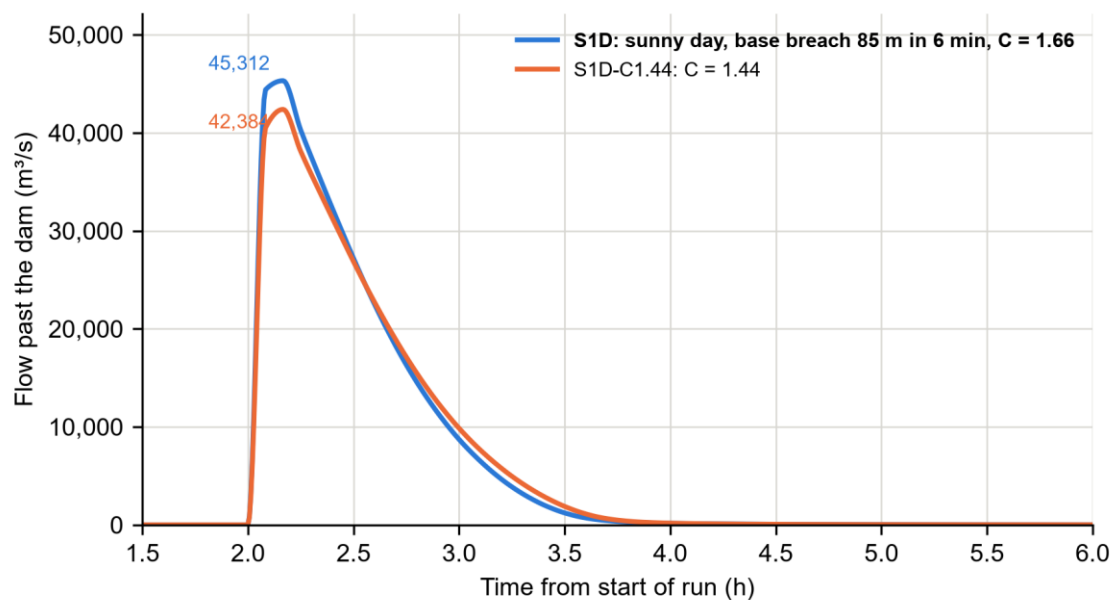


Figure 26 - Flow past the dam for the two breach weir coefficients, sunny day.

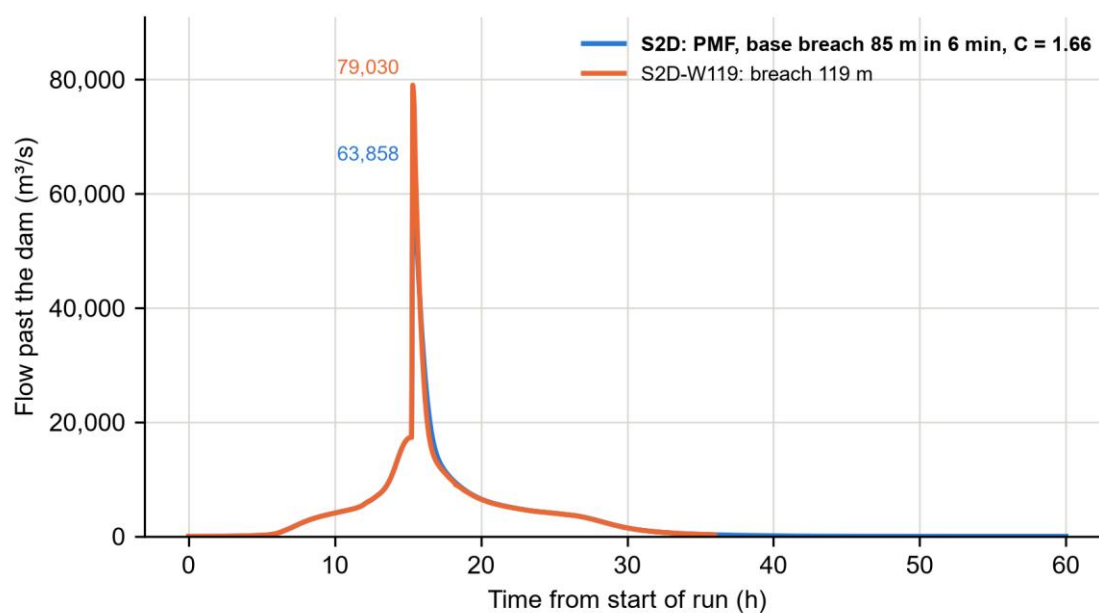


Figure 27 - Flow past the dam for the flood-day breach-width check (85 and 119 m).

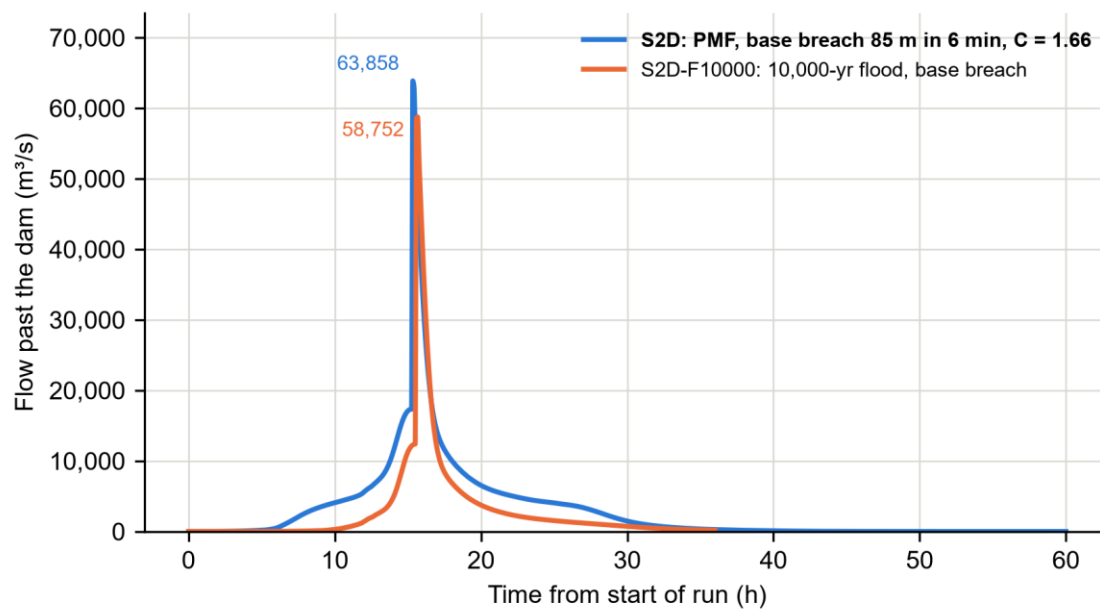


Figure 28 - Flow past the dam for the flood-day failure during the PMF and during the 10,000-year flood.



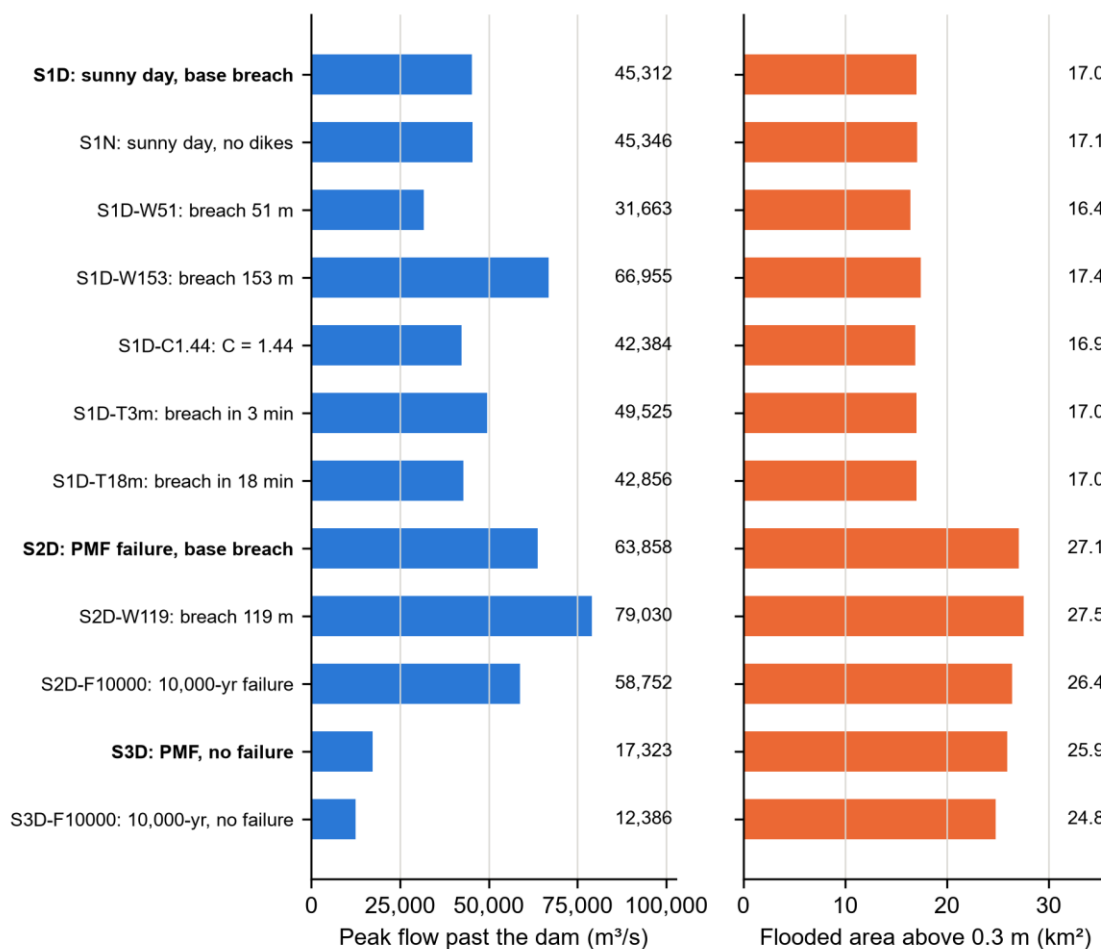


Figure 29 - Peak flow and flooded area of all runs.

In short: the breach width sets the peak almost in proportion (31,663, 45,312 and 66,955 m³/s for 51, 85 and 153 m) while the flooded area hardly moves (16.4 to 17.4 km²), because the plain is wide and flat. The formation time changes the peak by about a tenth either way (49,525 m³/s in 3 minutes, 42,856 in 18) and neither the extent nor the people in it, but it does change the first quarter hour: 18,050 people are reached within 15 minutes of a 3-minute breach, 15,579 with 6 minutes and 4,871 with 18. The lower weir coefficient trims the peak by 6 %. On the flood day a 119 m breach raises the peak from 63,858 to 79,030 m³/s and the flooded area by 0.5 km²; the 10,000-year flood gives the same picture as the PMF with about a tenth less flow.

9. CONSEQUENCES

People and land use in the flooded area were counted on the population grid and the plot layer of the flood-risk study, as set out in chapter 2. Table 19 gives, for every run, the people in the flooded area and how many of them are where the flood is unsafe for people (H4 or worse), the plots touched, and the area in each hazard class; Figure 30 draws the people by class.

Table 19 - People, plots and hazard classes per scenario.

Scenario	People in flooded area	of whom in H4 or worse	Plots touched	of which residential	H1 to H3 (km ²)	H4 (km ²)	H5 (km ²)	H6 (km ²)
S1D	23,212	22,541	2,950	1,677	0.9	0.9	6.8	8.7
S1N	23,299	22,655	2,963	1,687	0.9	0.9	6.7	8.9
S1D-W51	21,999	21,491	2,806	1,557	1.0	1.2	7.1	7.4
S1D-W153	24,148	23,819	3,036	1,756	0.8	0.8	5.6	10.5
S1D-C1.44	22,876	22,394	2,936	1,668	0.9	0.9	6.9	8.5
S1D-T3m	23,212	22,541	2,950	1,677	0.9	0.9	6.8	8.7
S1D-T18m	23,211	22,540	2,950	1,677	0.9	0.9	6.8	8.7
S2D	46,299	42,911	3,341	2,038	1.8	1.4	6.6	18.4
S2D-W119	46,960	43,451	3,404	2,099	1.9	1.3	6.6	19.0
S2D-F10000	43,376	40,641	3,293	1,992	2.0	2.1	6.1	17.4
S3D	44,088	40,121	3,065	1,777	2.2	1.9	10.1	13.0
S3D-F10000	39,083	35,537	2,780	1,572	2.8	2.7	12.1	8.7



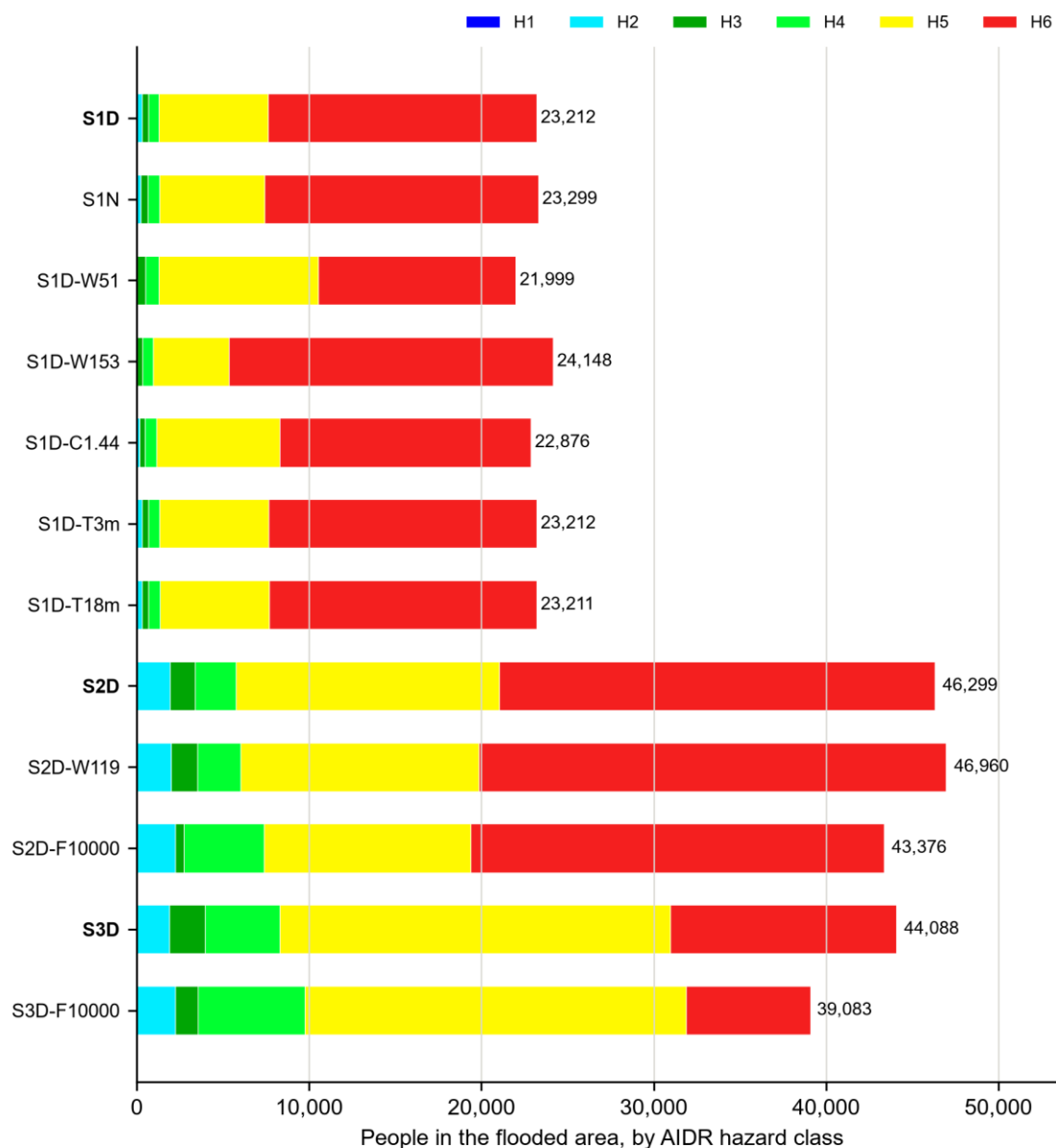


Figure 30 - People in the flooded area by hazard class, per scenario.

9.1. WARNING TIME: WHO IS REACHED, AND WHEN

The hazard maps alone make the flood without failure look almost as bad as the failure: similar area, similar classes. What separates them is time. The sunny-day wave reaches the town within minutes of the breach; the PMF takes hours to build up, and its rain and side wadis wet the plain long before the reservoir spills. This section puts the scenarios on the clock that matters for the emergency plan, with the definition of chapter 2: a failure is timed from the breach, the flood without failure from the start of the storm, and the flood-day failure by the extra depth its wave adds to the flood. The times



are travel times; the warning time available to people is shorter by the time it takes to detect the failure and to spread the alarm, which depend on the dam's instrumentation and the authorities' procedures and are not set by this study.

Table 20 gives the time at which the rise reaches five places along the wadi, from the gorge exit to the shore; the flood-day columns are the failure wave over the flood, the S3 columns count from the start of the storm.

Table 20 - Arrival of a 0.3 m rise at five places: failures timed from the breach, the flood without failure from the start of the storm.

Place	S1D	S1N	S2D	S2D-F10000	S3D	S3D-F10000
Gorge exit, start of the fan	5 min	5 min	< 5 min	< 5 min	40 min	1 h 45 min
Head of the training dikes	5 min	5 min	< 5 min	< 5 min	43 min	2 h 58 min
Qurayat, residential centre	15 min	10 min	10 min	10 min	1 h 06 min	3 h 15 min
Shoreline, main outlet	22 min	22 min	10 min	11 min	1 h 38 min	4 h 30 min
Shoreline, southern outlet	not reached	not reached	55 min	1 h 06 min	2 h 57 min	7 h 26 min

Table 21 counts, for the failures, the people and the plots reached within each time after the breach; the last column is everyone in the flooded area, whether or not the rise reaches 0.3 m. Figure 31 draws the same counts as curves.



Table 21 - People and plots reached within a given time after the breach (cumulative).

Scenario	within 15 min	within 30 min	within 45 min	within 1 h	within 1 h 30 min	within 2 h	within 3 h	Total
S1D, people	15,579	21,739	22,706	23,179	23,179	23,179	23,179	23,212
S1D, plots	1,944	2,493	2,831	2,949	2,950	2,950	2,950	2,950
S1N, people	15,765	21,822	22,899	23,266	23,267	23,267	23,267	23,299
S1N, plots	1,991	2,494	2,844	2,961	2,963	2,963	2,963	2,963
S1D-W51, people	7,586	20,715	20,945	21,740	21,966	21,966	21,966	21,999
S1D-W51, plots	1,267	2,368	2,480	2,726	2,802	2,802	2,802	2,802
S1D-W153, people	21,948	22,994	24,083	24,116	24,116	24,116	24,116	24,148
S1D-W153, plots	2,459	2,695	3,019	3,035	3,036	3,036	3,036	3,036
S1D-C1.44, people	13,866	21,427	22,251	22,795	22,844	22,844	22,844	22,876
S1D-C1.44, plots	1,842	2,481	2,776	2,930	2,936	2,936	2,936	2,936
S1D-T3m, people	18,050	21,786	22,845	23,179	23,179	23,179	23,179	23,212
S1D-T3m, plots	2,147	2,511	2,847	2,950	2,950	2,950	2,950	2,950
S1D-T18m, people	4,871	21,697	22,176	23,108	23,178	23,178	23,178	23,211
S1D-T18m, plots	693	2,463	2,661	2,913	2,950	2,950	2,950	2,950
S2D, people	25,365	26,129	26,887	31,500	31,631	31,936	31,936	46,299
S2D, plots	3,250	3,331	3,341	3,341	3,341	3,341	3,341	3,341
S2D-W119, people	25,787	26,787	30,089	33,466	33,746	34,175	34,175	46,960
S2D-W119, plots	3,305	3,403	3,404	3,404	3,404	3,404	3,404	3,404
S2D-F10000, people	24,293	25,515	26,126	26,814	30,757	30,757	30,757	43,376
S2D-F10000, plots	3,074	3,248	3,292	3,293	3,293	3,293	3,293	3,293

Table 22 does the same for the flood without failure, in hours from the start of the storm.

Table 22 - People and plots reached within a given time after the start of the storm, no failure (cumulative).

Scenario	within 3 h	within 6 h	within 9 h	within 12 h	within 15 h	within 18 h	Total
S3D, people	11,921	27,933	38,020	44,017	44,056	44,056	44,088
S3D, plots	1,258	2,200	2,754	3,057	3,065	3,065	3,065
S3D-F10000, people	359	5,682	18,314	37,473	39,051	39,051	39,083
S3D-F10000, plots	26	704	1,600	2,702	2,780	2,780	2,780



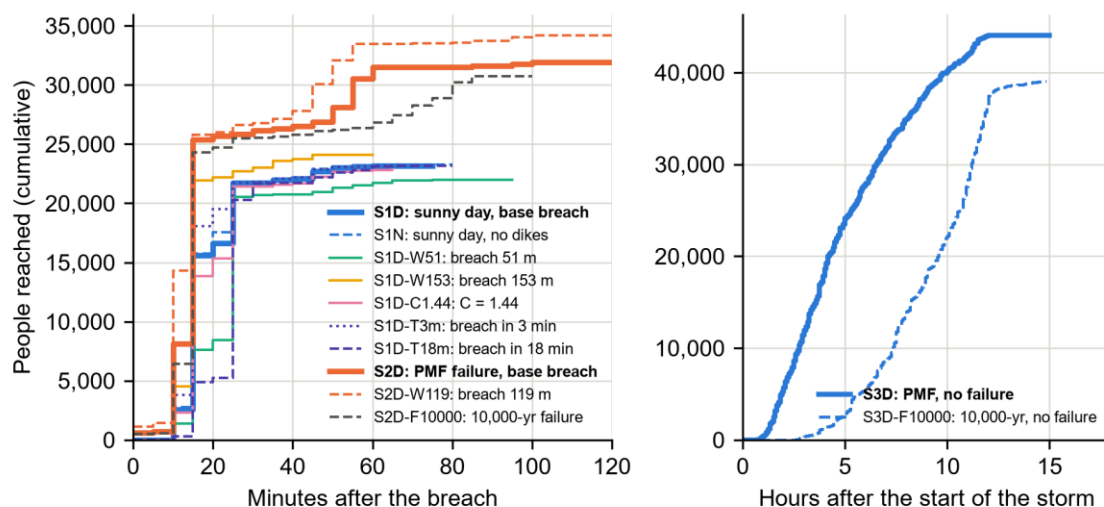


Figure 31 - People reached against time: failures from the breach (left), the flood without failure from the start of the storm (right).

Table 23 splits the people reached in each interval by the hazard class they end up in, the two inputs a life-safety estimate needs: how little warning, and how severe the flood. H4 and worse is unsafe for people on foot.

Table 23 - People newly reached in each interval after the breach, by final hazard class: sunny-day and flood-day failure.

Reached within	S1D H1-H3	S1D H4	S1D H5	S1D H6	S2D H1-H3	S2D H4	S2D H5	S2D H6
15 min	52	53	3,242	12,233	101	117	3,411	21,736
15 min to 30 min	161	215	2,447	3,337	362	116	276	11
30 min to 45 min	142	186	639	0	191	92	347	128
45 min to 1 h	315	158	0	0	10	31	3,347	1,226
1 h to 1 h 30 min	0	0	0	0	0	0	127	5
1 h 30 min to 2 h	0	0	0	0	0	0	98	206

Figure 32 to Figure 34 show where the people are: the population grid with the flooded area of each scenario outlined.



S1D: people in the flooded area

S1D sunny day failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

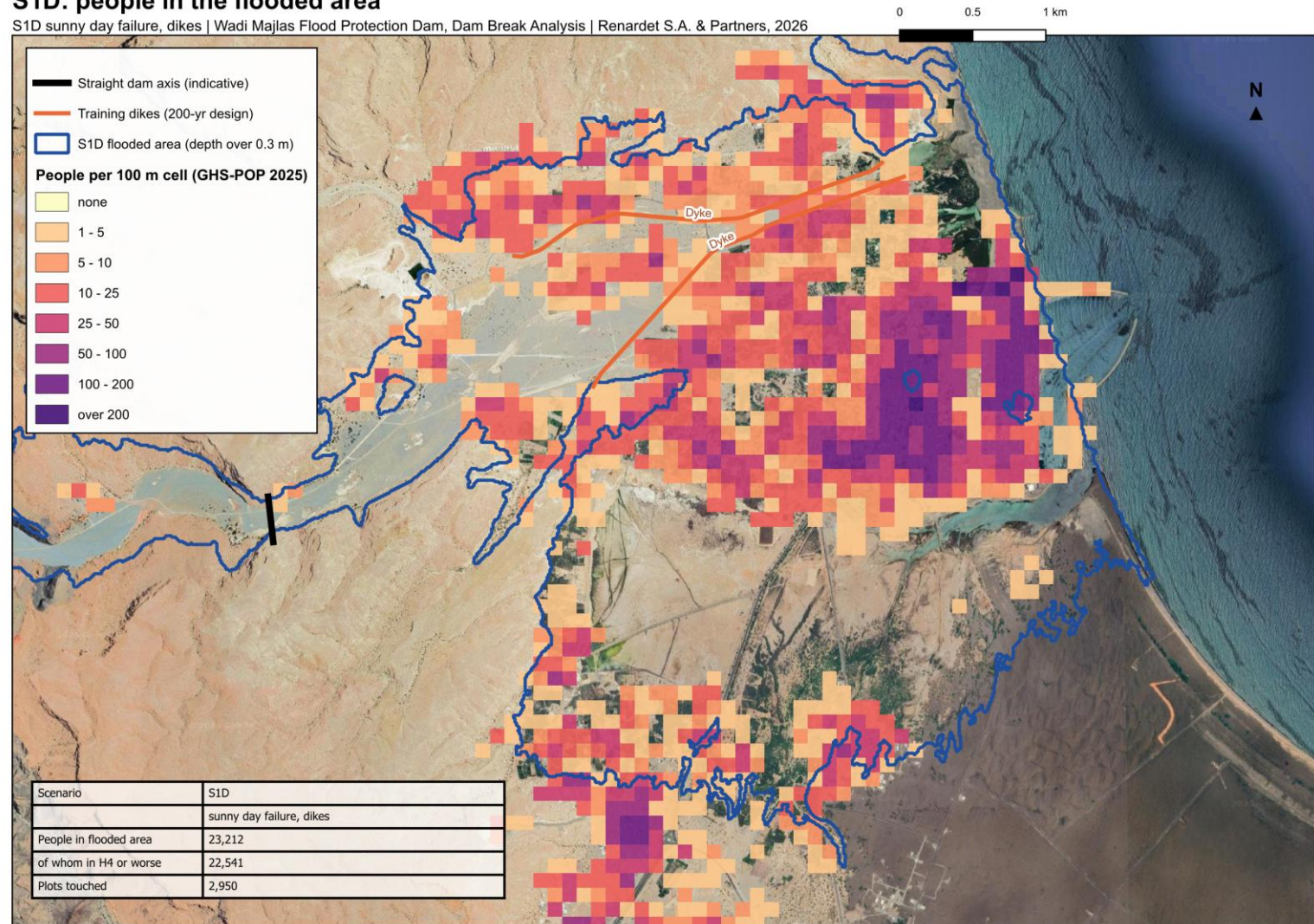


Figure 32 - S1D: people per 100 m cell with the flooded area outlined.

S2D: people in the flooded area

S2D flood day PMF failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

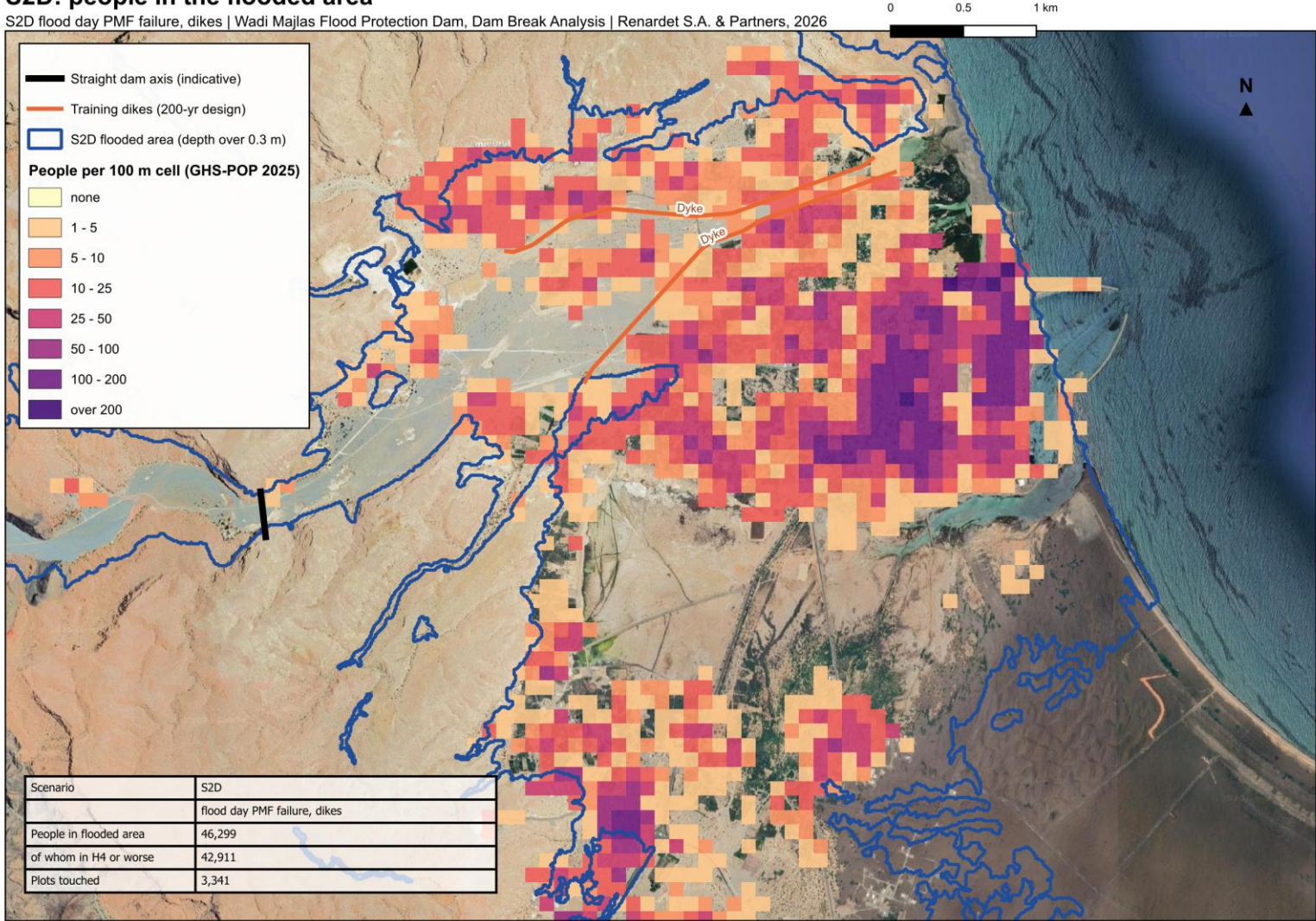


Figure 33 - S2D: people per 100 m cell with the flooded area outlined.



S3D: people in the flooded area

S3D PMF no failure, dikes | Wadi Majlas Flood Protection Dam, Dam Break Analysis | Renardet S.A. & Partners, 2026

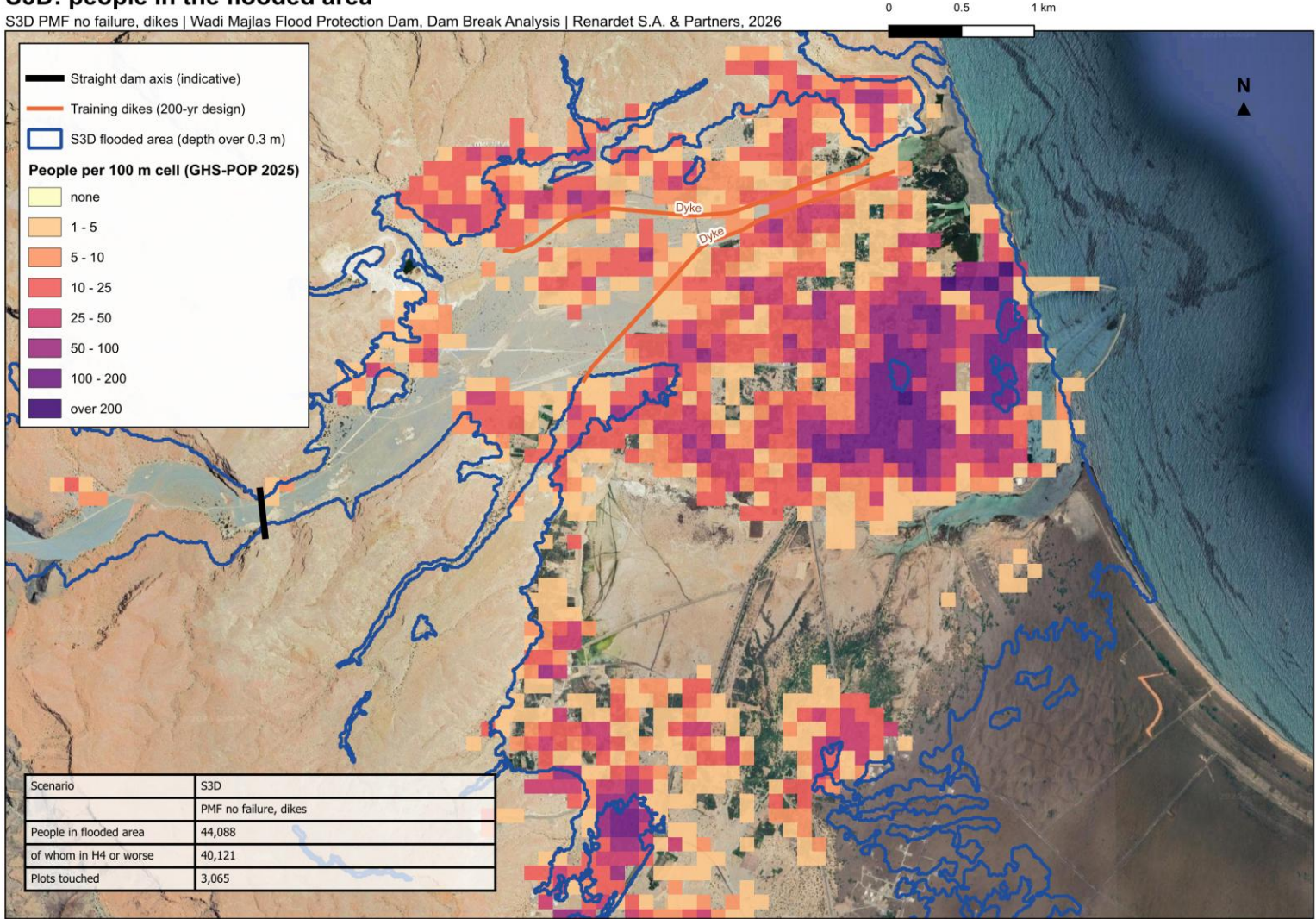


Figure 34 - S3D: people per 100 m cell with the flooded area outlined.



10. DISCUSSION

10.1. WHAT THE FAILURE ADDS TO THE FLOOD

The flood-day runs are read against the no-failure twins: the difference in depth, arrival time and hazard class is the effect of the dam, and only that difference belongs to the dam-safety classification. The sunny-day runs stand alone: there is no flood to compare with, and everything in the flooded area is the dam's doing. Measured by extent the failure adds little to the PMF, because the flood has already covered the plain; measured by depth and hazard class it adds a third more area in the worst class; measured by time it changes everything, because the flood gives hours and the failure wave gives minutes. Arrival time is the most important difference between the scenarios.

10.2. THE TRAINING DIKES

The dikes were designed to pass the 200-year wadi flood. A dam break wave is one to two orders of magnitude larger and overtops them within minutes. The comparison of the sunny-day failure with and without them showed no significant difference in flooded area, people at risk or arrival time, so the analysis was carried on with the dikes in place only. The plan should not rely on them.

10.3. LIMITS OF THE ANALYSIS

- The breach is prescribed, not computed: its width and speed follow the guidance for concrete gravity dams (chapter 2) and are bounded by the sensitivity runs.
- The terrain of the plain is the 5 m digital terrain model; buildings are not resolved, so depths in the town are those of an open plain with the town's roughness.
- The population grid is a 2025 estimate at 100 m; the plot layer is the flood-risk study layer.
- The sea level is fixed at mean higher high water, 0.71 m a.s.l.; a storm surge coinciding with the PMF was not modelled.
- The arrival times are travel times of the water; detection and alarm times are not included.



11. CONCLUSIONS AND RECOMMENDATIONS

11.1. WHAT EACH SCENARIO MEANS

Sunny-day failure (S1D). A failure of five monoliths with the reservoir full releases 103 Mm^3 with a peak of $45,312 \text{ m}^3/\text{s}$ and floods 17.0 km^2 of the coastal plain, 8.7 km^2 of it in the worst hazard class, where buildings fail. About 23,212 people live in that area and 22,541 of them where the water is unsafe for people. The wave reaches the residential centre 15 min after the breach and the shoreline 22 min after; 15,579 people are reached within 15 minutes and 23,179 within an hour. This is the scenario the emergency plan must be written for: there is no flood to warn of, so the warning has to come from the dam's own monitoring, before the failure.

Sunny-day failure without the dikes (S1N). The result is the same as with them: 17.1 km^2 and 23,299 people against 17.0 km^2 and 23,212. That is why the other scenarios were run with the dikes in place only.

Flood-day failure (S2D). A failure at the peak of the PMF raises the peak flow from 17,323 to $63,858 \text{ m}^3/\text{s}$ but the flooded area only from 25.9 to 27.1 km^2 , because the plain is already under the flood. The failure adds depth, force and above all speed: the worst hazard class grows from 13.0 to 18.4 km^2 , the people in the flooded area from 44,088 to 46,299, and the failure wave adds 0.3 m or more to the water standing on 25,365 people within 15 minutes of the breach, reaching the shoreline in 10 min. The 10,000-year flood gives the same picture with a tenth less flow.

PMF without failure (S3D). The spillway passes the PMF with a maximum pool of 96.06 m a.s.l. , 0.64 m below the parapet: the dam is not overtopped. The flood alone covers 25.9 km^2 with 44,088 people, 13.0 km^2 of it in the worst class, but it takes 1 h 06 min to reach the residential centre and about twelve hours to reach everyone: there is time to warn and to move people, which the failure scenarios do not give.

Figure 35 puts the three scenarios side by side on the four measures that matter: how much water, how much of the plain and how severely, how many people, and how fast they are reached.



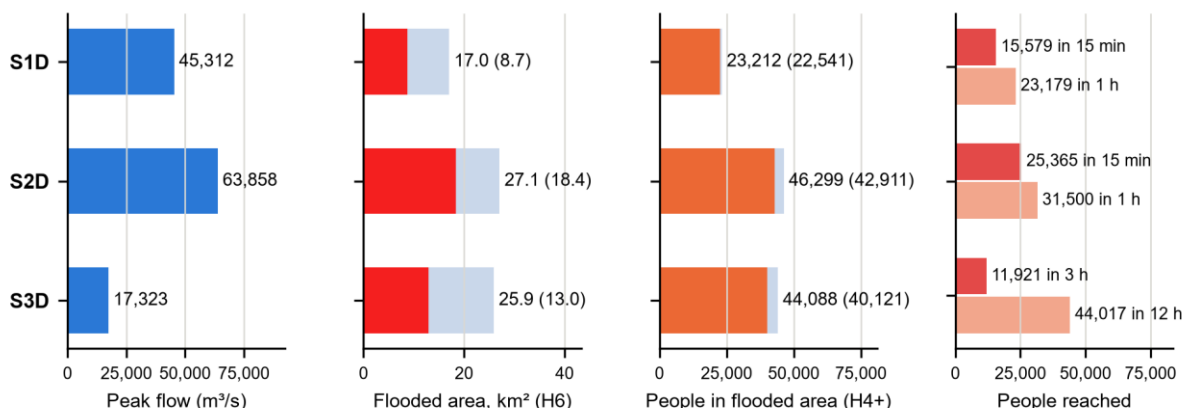


Figure 35 - The three scenarios compared: peak flow, flooded area with the H6 share, people in the flooded area with the H4+ share, and people reached early and late.

11.2. SENSITIVITY

Table 24 - What the sensitivity runs move, from the lowest to the highest value tested.

Parameter varied	Range tested	Peak flow at the dam	Flooded area	People reached within 15 min
Breach width, sunny day	51, 85, 153 m (3, 5, 9 monoliths)	31,663 to 66,955 m³/s	16.4 to 17.4 km²	7,586 to 21,948
Formation time, sunny day	3, 6, 18 min	42,856 to 49,525 m³/s	17.0 to 17.0 km²	4,871 to 18,050
Weir coefficient, sunny day	1.44, 1.66	42,384 to 45,312 m³/s	16.9 to 17.0 km²	13,866 to 15,579
Breach width, flood day	85, 119 m (5, 7 monoliths)	63,858 to 79,030 m³/s	27.1 to 27.5 km²	25,365 to 25,787
Flood, with failure	10,000-year, PMF	58,752 to 63,858 m³/s	26.4 to 27.1 km²	24,293 to 25,365
Flood, without failure	10,000-year, PMF	12,386 to 17,323 m³/s	24.8 to 25.9 km²	-

The breach width is the parameter that matters for the peak flow, and the peak follows it almost in proportion; the flooded area and the people in it hardly move with any parameter, because the plain is wide and flat and fills either way. The formation time matters for the first quarter hour of warning and for nothing else. The base case therefore gives a robust picture, and the bounds of Table 24 should be quoted with it.

11.3. RECOMMENDATIONS

- The emergency action plan should be built on the sunny-day failure: its arrival isochrones are the warning zones, and the hazard maps give the evacuation routes and the safe ground.



- Warning must come from the dam, before the failure: instrumentation and an alarm procedure with a detection-plus-dissemination time well under fifteen minutes, since that is how long the wave takes to reach the town.
- During a major flood the plain is already flooded and the failure wave arrives within minutes; the evacuation of the low plain should therefore be part of the flood-warning procedure itself, not wait for a dam alarm.
- The training dikes should not be relied on in the plan: they are overtopped by every failure scenario.
- The results should be quoted with their bounds: peak flow within the range of the breach widths tested, the flooded area and the people in it essentially fixed.



REFERENCES

- AIDR (2017). Guideline 7-3: Flood Hazard. Australian Institute for Disaster Resilience, Melbourne.
- Brunner, G.W. (2014). Using HEC-RAS for Dam Break Studies. Training Document TD-39, Hydrologic Engineering Center, US Army Corps of Engineers, Davis, California.
- FERC (2015). Engineering Guidelines for the Evaluation of Hydropower Projects, Chapter II: Selecting and Accommodating Inflow Design Floods for Dams. Federal Energy Regulatory Commission, Washington DC.
- Froehlich, D.C. (1995). Embankment dam breach parameters revisited. Water Resources Engineering, Proceedings of the 1995 ASCE Conference, 887-891.
- Froehlich, D.C. (2008). Embankment dam breach parameters and their uncertainties. Journal of Hydraulic Engineering, 134(12), 1708-1721.
- ICOLD (2020). Bulletin 111: Dam Break Flood Analysis, Review and Recommendations. International Commission on Large Dams, Paris.
- MacDonald, T.C. and Langridge-Monopolis, J. (1984). Breaching characteristics of dam failures. Journal of Hydraulic Engineering, 110(5), 567-586.
- Smith, G.P., Davey, E.K. and Cox, R.J. (2014). Flood Hazard. Technical Report 2014/07, Water Research Laboratory, University of New South Wales.
- US Army Corps of Engineers (2024). HEC-RAS River Analysis System, version 6.6: User's Manual, 2D Modeling User's Manual and Hydraulic Reference Manual. Hydrologic Engineering Center, Davis, California.
- Von Thun, J.L. and Gillette, D.R. (1990). Guidance on breach parameters. Internal memorandum, US Bureau of Reclamation, Denver.
- Xu, Y. and Zhang, L.M. (2009). Breaching parameters for earth and rockfill dams. Journal of Geotechnical and Geoenvironmental Engineering, 135(12), 1957-1970.



APPENDIX A: RUN REGISTER

Fourteen runs were made for this report: two reservoir fills that produced the starting state of each geometry, the four scenario runs and the eight sensitivity runs, about 51 hours of computation in all. Table 25 lists them with their simulation window and the solver's own completion status.

Table 25 - Run register.

Run	Description	Simulation window	Solver status
Fill	reservoir fill, geometry with dikes	48 h	restart file written
FillN	reservoir fill, geometry without dikes	48 h	restart file written
S1D	sunny-day failure, dikes in place	14 h	completed
S1N	sunny-day failure, no dikes	14 h	completed
S1D-W51	sunny-day failure, breach 51 m wide (three monoliths)	14 h	completed
S1D-W153	sunny-day failure, breach 153 m wide (nine monoliths)	14 h	completed
S1D-C1.44	sunny-day failure, breach weir coefficient 1.44	14 h	completed
S1D-T3m	sunny-day failure, breach formed in 3 minutes	14 h	completed
S1D-T18m	sunny-day failure, breach formed in 18 minutes	14 h	completed
S2D	flood-day (PMF) failure, dikes in place	60 h	completed
S2D-W119	flood-day failure, breach 119 m wide (seven monoliths)	36 h	completed
S2D-F10000	flood-day failure during the 10,000-year flood	36 h	completed
S3D	PMF without failure, dikes in place	60 h	completed
S3D-F10000	10,000-year flood without failure, dikes in place	36 h	completed

